

Local Inhibitory Dynamics Underpin Temporal Integration and Functional Segregation between Barrels and Septa in the Mouse Barrel Cortex

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Argunşah et al. investigate the mechanisms underlying the differential response dynamics of barrel vs septa domains in shaping the responses to single vs multiple whiskers. Based on the observation of a higher density of SST+ interneurons in the septa, the authors investigate the hypothesis that Elfn1-dependent short-term plasticity shapes these responses. This **important** study is, however, supported by **incomplete** evidence; factors restricting the strength of evidence are the limited spatial resolution of the multi-unit activity, as well as the lack of a mechanistic explanation. This provocative and intellectually stimulating hypothesis provides a contribution to work on how different cell types shape cortical representation.

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Abstract

Just as separated digits and repeated sampling enhance somatosensation in humans, mice sense objects through multiple segregated whiskers through successive contacts. Individual whisker identity is maintained through the somatotopic organization of the Whisker → Brainstem → Thalamus → Cortex axis, culminating in distinct cortical domains: barrels and the surrounding septa. By performing simultaneous recordings using in-vivo electrophysiology in wild-type (WT) mice, we identify a progressive divergence in spiking activity between these domains upon repeated behaviorally relevant (10Hz) single- and multi-whisker stimulation. While the spiking activity ratio of multi- to single-whisker stimulation remains stable in barrels, it increases progressively in septa, suggesting inhibitory cell recruitment. Using genetic fate-mapping and tissue clearing, we indeed reveal that SST+ and VIP+ interneurons exhibit distinct laminar and regional distributions in barrel and septa domains. Further, calcium imaging of SST+ and VIP+ interneurons shows that

while both neuron types respond to single-whisker stimulus, SST+ interneurons preferentially engage more in 10Hz multi-whisker stimulation, indicating their critical role in progressive stimulus preference. Genetic removal of *Elfn1*, which regulates the incoming excitatory synaptic dynamics onto SST+ interneurons, leads to the loss of the progressive increase in septal spiking activity upon multi-whisker stimulation, resembling the stable response pattern observed in barrels. The importance of the loss of functional segregation of barrels, versus septa is revealed by cumulative temporal decoding analysis, supporting the notion that SST+ interneuron-mediated inhibition contributes to temporal encoding and stimulus integration. Finally, viral tracing combined with whole brain clearing and imaging reveals that barrel and septa domains project differentially to secondary somatosensory (S2) and motor (M1) cortices. These distinct projection patterns suggest that differential inhibitory processing in barrels and septa may contribute to functionally specialized downstream signaling. Together, our findings indicate that the progressive engagement of SST+ interneurons, mediated by *Elfn1*-dependent synaptic facilitation, underlies the preferential integration of multi-whisker stimuli in septa. This local inhibitory mechanism likely contributes to the functional segregation of barrel and septa domains and their distinct cortical projections, shaping how sensory information is processed and relayed to higher-order brain regions.

Introduction

Although the somatosensory whisker system is one of the principal means by which mice sense their environment and navigate the world, these tasks are carried out by a relatively small number of whiskers. This small number of vibrissae is represented in the whisker somatosensory cortex as barrel “islands” (columns) separated and surrounded by the septal “sea” (compartments), reminiscent of the whisker pad pattern on a rodent’s face, where sparse hair follicles are separated by spaces between them. In contrast, other sensory systems, such as the auditory and the visual ones, have densely packed anatomical receptor configurations and cortical representations. Although all sensory cortices are composed of canonical microcircuits (Douglas and Martin, 2004) that include similar populations of excitatory and inhibitory neurons, maintaining segregated individual whisker information may require anatomical specializations, the barrel and septa domains. These domains may therefore create a distinct spatiotemporal stimulus representation and information coding in the barrel cortex, different from the other sensory modalities. In the adult murine barrel cortex, thalamocortical inputs coming from the ventro-postero-medial (VPM) and postero-medial (PoM) nuclei segregate into the barrel and septa domains, respectively (Alloway, 2008; Bureau et al., 2006; Kim and Ebner, 1999; Staiger and Petersen, 2021). Further, in mice (Audette et al., 2018; Sato and Svoboda, 2010; Shepherd and Svoboda, 2005) and rats (Alloway, 2008; Chakrabarti and Alloway, 2006; Melzer et al., 2006b) it was shown that information processing and flow within and from these domains target anatomically distinct sensory and motor areas. It has been suggested that the primary whisker somatosensory cortex (wS1) sends higher dimensional information needed for object recognition to the secondary somatosensory cortex (wS2), while the projections to the primary motor cortex (M1) carry less complex information that does not require segregated streams of information coming from individual whiskers (Alloway, 2008; Brecht and Sakmann, 2002; Cai et al., 2022; Sato and Svoboda, 2010). This divergence in information processing is also found within the barrel- and septa-related circuits. While barrel circuits are more involved in processing spatiotemporal whisker-object interactions, the septal circuits are more sensitive to the frequency of whisker movements, which suggests the use of temporal versus rate coding of information in barrel and septa, respectively (Melzer et al., 2006a, 2006b). Hence the existence of the septal column separating the barrels may provide mice with the means to separate these two interconnected yet very distinct information-processing routes along the temporal domain, and to make sense of their somatosensory environment through repeated sampling via cloistered information streams. Recent studies in mice have shown that barrel and septa compartments are

not only anatomically segregated in terms of cell-type specific thalamocortical projections and differential expression of different genetic markers (Young et al., 2023 [DOI](#)) but also in their tuning upon single- and multi- whisker stimulation (Wang et al. 2022 [DOI](#)).

In both rats and mice, the operational identity of barrels and septa has often been attributed to the parallel bottom-up thalamocortical pathways of VPM and PoM. Here, by performing *in-vivo* silicon probe recordings simultaneously in the barrel and septa domains upon repeated single- and multi-whisker stimulation in wS1 of mice, we first show and characterize the temporal divergence between these domains. Utilizing genetic fate-mapping, passive tissue clearing, and light-sheet microscopy, we reveal that cortical somatostatin (SST+) and vasoactive intestinal peptide (VIP+) expressing interneurons show a layer-dependent differential distribution in the barrel versus septa columns. Through two-photon calcium imaging of these two populations, we show a differential engagement of SST+ cells upon multi- vs. single-whisker stimulation. By altering the short-term synaptic dynamics of incoming excitation onto these interneurons through the removal of the trans-synaptic protein Eln1 (Eln1 KO), we find that the column-specific contrast ratio between multi- over single- whisker response to a repetitive stimulus is diminished. Through a progressive decoding analysis, we find that in contrast to control, whisker stimulation-evoked Eln1 KO responses in different domains cannot be decoded efficiently. Finally, using *in-vivo* retrograde viral-based tracing, we show a layer-specific projection preference between these domains and downstream regions wS2 and M1. Hence, in addition to the anatomical specialization through thalamocortical pathways, we reveal a key contribution of local lateral cortical inhibition, provided by SST+ interneurons, in setting up the functional segregation of barrel and septa domains, and subsequently, their downstream targets.

Results

Barrel and Septa columns display differentially adaptive spiking upon repeated single- and multi- whisker stimulation

To characterize the temporal stimulus-response profiles of barrel and septal domains simultaneously, we performed *in-vivo* multi-shank (8×8) silicon probe recordings under urethane-induced light anesthesia from mouse somatosensory whisker barrel cortex (wS1), after the age of active whisking onset, postnatal day (P)20-30. Whisker stimulation was performed on either just one principal whisker (single-whisker stimulus or SWS) or via both the principal whisker and most of the macro vibrissae together (multi-whisker stimulus or MWS; see methods) (Figure 1A [DOI](#)). Single- or multi- whisker evoked activity within identified regions was recorded upon a 2s-long stimulation at 10Hz, similar to the average frequency at which mice actively whisk (Carvell and Simons, 1990 [DOI](#); Grant et al., 2012 [DOI](#)). Each mouse received both stimulation paradigms. Current source density (CSD) analysis was used to assess the location of each silicon probe shank, which was later verified by post-hoc histology. The recorded activity was assigned to the stimulated principal barrel column (B), the adjacent septal column (S), or the adjacent unstimulated neighboring barrel column (N) according to SWS paradigm (Figure 1B [DOI](#) and Supp Figure 1A [DOI](#) to D [DOI](#)). An example probe assignment can be seen in Supp Figure 1E [DOI](#). While this distinctive experimental paradigm is relatively low throughput, it has allowed us to investigate all three of these domains simultaneously. The septal compartments in mouse barrel cortex are quite narrow (Bureau et al., 2006 [DOI](#); Sato et al., 2007 [DOI](#)) and there is the possibility of rotational misalignment during the insertion of silicon probes into the brain (Supp. Figure 1F [DOI](#)). For these reasons we restricted our analysis of *in-vivo* responses upon SWS and MWS between barrel and septa responses to upper layers and L4, for which we have higher confidence about the anatomical domain allocation after our histological (Supp. Figure 1B [DOI](#)) and CSD analysis (Supp. Figure 1C [DOI](#) and D [DOI](#)). In line with this domain allocation, t-distributed stochastic neighbor embedding (t-SNE) analysis of the multi-unit activity (MUA) recorded across the entire train of stimuli in the different

domains revealed that the spiking responses between the principal barrel and the neighboring one form two separate clusters, while the responses recorded in the septa lay in between (**Figure 1C**; see methods).

It has previously been shown that electrophysiological responses upon whisking-based object detection saturate rapidly after a single whisker touch, while object localization may require multiple repeated whisker touches (O'Connor et al., 2013; Pammer et al., 2013). Correspondingly, when a single whisker is stimulated repeatedly, the response to the first pulse is principally bottom-up thalamic-driven responses, while the later pulses in the train are expected to also gradually engage cortico-thalamo-cortical and cortico-cortical loops. We therefore first tested whether the leading pulses of the stimulus train evoked differential responses among the principal barrel, the adjacent septa, and the neighboring barrel upon SWS and MWS. For easier visualization and full appreciation of the data, we present our findings both in classical time series firing-rate plot and stimulation pulse aligned stacked images (**Supp. Figure 2A** and **B**, 200ms baseline followed by 2000ms evoked response to 20 pulses at 10Hz reshaped to 20 stacked 100ms-long evoked response image, left to right). Earlier studies that separately recorded whisker-evoked responses from either barrel or septa domains in rat reported that the responses of barrel neurons to single principal whisker stimulation are much higher than those evoked in the septal neurons (Armstrong-James and Fox, 1987; Brecht and Sakmann, 2002). The analysis of the data we obtained from the simultaneous recordings at these domains in the mouse do not reveal such differences in the first two stimulus responses in L4 upon single whisker stimulation (SWS) (**Figure 1D**; statistics on the bottom; white represents no significance). However, response differences emerged from the 3rd pulse onward upon SWS (**Figure 1D**, bottom). Similar differences also emerged between stimulated principal and unstimulated neighboring barrel columns upon SWS (starting from 4th pulse, **Figure 1D**, bottom, Barrel-vs-Neighbor (BN) row). Interestingly, an inverted pattern of significance appeared between septa and adjacent unstimulated neighboring barrels (**Figure 1D**, bottom, Septa-vs-Neighbor (SN) row). When multiple whiskers (MWS) were activated simultaneously, like SWS, a divergence between barrel and septa domain activity also occurred in Layer 4 from the 2nd pulse onward (**Figure 1E**). Finally, when we calculated the response ratio upon MW over SW stimulation, we saw that the septal MW/SW ratio diverged from barrel MW/SW ratio over the course of repeated stimulation (**Figure 1F**), indicating a role of septal neurons in the progressive processing MW information.

Barrel and Septa columns display differential SST+ and VIP+ neuron densities

It is known that barrel and septa domains receive different thalamic projections, from VPM and PoM respectively, that might affect stimulus response properties (Ahissar et al., 2001; Chmielowska et al., 1989; Koralek et al., 1988; Sosnik et al., 2001). Our electrophysiological experiments show a significant divergence of responses over time upon both SWS and MWS in L4 between barrels (principal and neighboring) and adjacent septa, with minimal initial difference. We hypothesized that this late divergence might be driven by differences in the local circuitry, and specifically inhibitory cells. We therefore next assessed the spatial distribution of two inhibitory cell populations that could provide distinct regulation of cortical activity in the temporal domain: SST+ and VIP+ cells.

Earlier research examining the three largest inhibitory neuron populations in the mouse barrel cortex showed that, while parvalbumin positive (PV+) cells do not show any differential density preference between barrel and septa columns, SST+ and VIP+ cells have higher density in the septa in L4 (Almási et al., 2019). We attempted to verify the reported differential SST+ and VIP+ neuronal distributions using an alternative imaging and counting strategy. To label the two neuronal populations, we crossed the SST-Cre and VIP-Cre lines with a tdTomato reporter mouse line (Ai14). To comprehensively quantify the density of SST+ and VIP+ cells in the barrel and septa domains in 3D, we utilized a passive CLARITY-based tissue clearing protocol, followed by light-

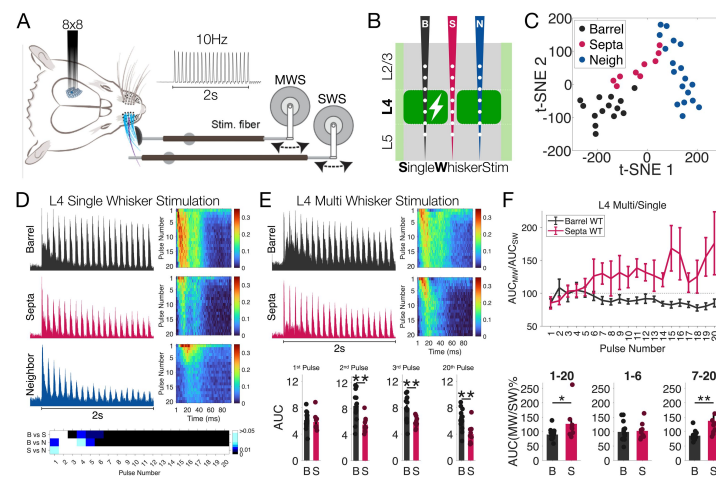


Figure 1.

Temporal divergences emerge between barrel and septa domains upon repeated whisker stimulation in Layer 4 of mouse wS1.

(A) Either principal whiskers (SWS, or single whisker) or most of the principal whiskers (MWS, or multi whisker) were stimulated for 2s at 10Hz (20 pulses) for 20 times (20 trials). (B) Acute simultaneous *in-vivo* silicon probe (8x8) recordings were performed from barrel and septa domains after the start of active whisking (>P21) (C) t-SNE analysis of the average firing dynamics from three domains after averaging 20 single trials. (D) Left: Average multi-unit firing profiles recorded at Layer 4 of barrel (black), unstimulated neighbor (blue) and septa (pink) columns upon 2s-long 10Hz repeated single whisker stimulation. Middle: Matrix representation of the data. Right: Pair-wise statistical comparisons of first 50ms of the responses (yellow line represents the cut off) of three conditions for all pulses (Mann-Whitney-U-Test). (E) Same with D but response profiles upon MW stimulation. (F) Top: Temporal ratio dynamics (average AUC MW responses divided by average AUC SW responses) for barrel and septa. Bottom: statistical comparison of average AUC ratios of all pulses (1-20), first six (1-6) or last 14 pulses (7-20), respectively. (p values are color coded in D, stars represents *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$ in E and F.)

sheet microscopy using a custom-built light-sheet microscope (Vladimirov et al., 2023; Voigt et al., 2019) (Figure 2A). The individual barrels in wS1 were reliably detected using auto-fluorescence from the tissue acquired with a 488nm laser. tdTomato positive SST+ or VIP+ neurons were detected by a 561nm laser, allowing us to accurately localize and count cells in barrel and septal domains (Figure 2B and C). By adjusting the angle of each barrel column in 3D (using Imaris, RRID:SCR_007370), we precisely identified the column through L2/3, L4 and L5, using the barrel borders for every plane (XY, YZ, XZ) in L4 (Figure 2B, also see Methods). Once the columns were accurately determined, we counted the number of tdTomato positive neurons in the barrels and septa and corresponding volumes for density estimation. Despite their opposite distributions and neuronal density changes along the depth of wS1 (Figure 2B and C), we found that both SST+ (Figure 2D) and VIP+ (Figure 2E) neuronal densities were higher in L4 septa compared to barrels, but we detected no difference in L2/3 and L5.

These results confirmed that SST+ and VIP+ interneurons have higher densities in septa compared to barrels in L4 and suggest these interneurons may play a role in diverging responses between barrels and septa upon SWS.

SST+ neurons are more strongly activated by multi- versus single-whisker stimulation

Having identified spatial distribution patterns of SST+ or VIP+ interneurons that could contribute to the single- vs multi- whisker response scaling observed in Figure 1F; we sought to assess these interneurons' response profiles upon similar whisker stimulation paradigms. Therefore, we performed similar SWS and MWS experiments using acute in- vivo two-photon calcium (Ca²⁺) imaging under light anesthesia after weaning (P21–41) (Figure 3A and B). We injected OGB-1, the membrane-permeable AM-ester form of the calcium indicator, into wS1 of animals expressing tdTomato in either VIP+ or SST+ INs (VIPCre-Ai14 and SSTCre-Ai14 lines, respectively) to localize VIP+ and SST+ activity (Figure 3A). Two-photon Ca²⁺ imaging was performed upon stimulation of either the principal C2 whisker at 10Hz for 2s (SWS) or most of the macro vibrissae including C2 (MWS) (Figure 3B). We found that both SST+ and VIP+ interneurons increased their firing upon SWS (Figure 3C and D) as has been reported previously (Kastli et al., 2020), but only SST+ interneurons changed their firing significantly between single- and multi- whisker stimulation at 10-Hz (Figure 3D, SW vs MW comparison). Although these optical recordings were performed in layer 2/3 (L2/3) rather than layer 4 (L4) due to depth limitations, the similarity of the peak MW/SW ratios of L2/3 populations in putative barrel and septa columns (c.f. Figure 1F and Supp. Figure 3) provides support for the validity of using L2/3 data to infer respective L4 cell activity. This result suggests that differential activation of SST+ interneurons is more likely to be involved in the domain specific temporal ratio differences between barrels and septa.

Progressive multi- over single-whisker response divergence in septal populations is abolished in Elfn1 KO mice

SST+ interneurons in the cortex are known to show distinct short-term synaptic plasticity, particularly strong facilitation of excitatory inputs, which enables them to regulate the temporal dynamics of cortical circuits. This facilitation allows SST+ cells to progressively enhance their inhibitory influence in response to repeated stimulation, a property critical for shaping network activity. A key regulator of this plasticity is the synaptic protein Elfn1, which mediates short-term synaptic facilitation of excitation on SST+ interneurons (Stachniak et al., 2021, 2019; Tomioka et al., 2014).

Having identified SST+ cells as potential regulators of temporal MW/SW ratio divergence, we investigated how the absence of Elfn1 alters sensory-driven activity in vivo, potentially shaping septal function in MW/SW response scaling. Using the same whisker stimulation paradigms and

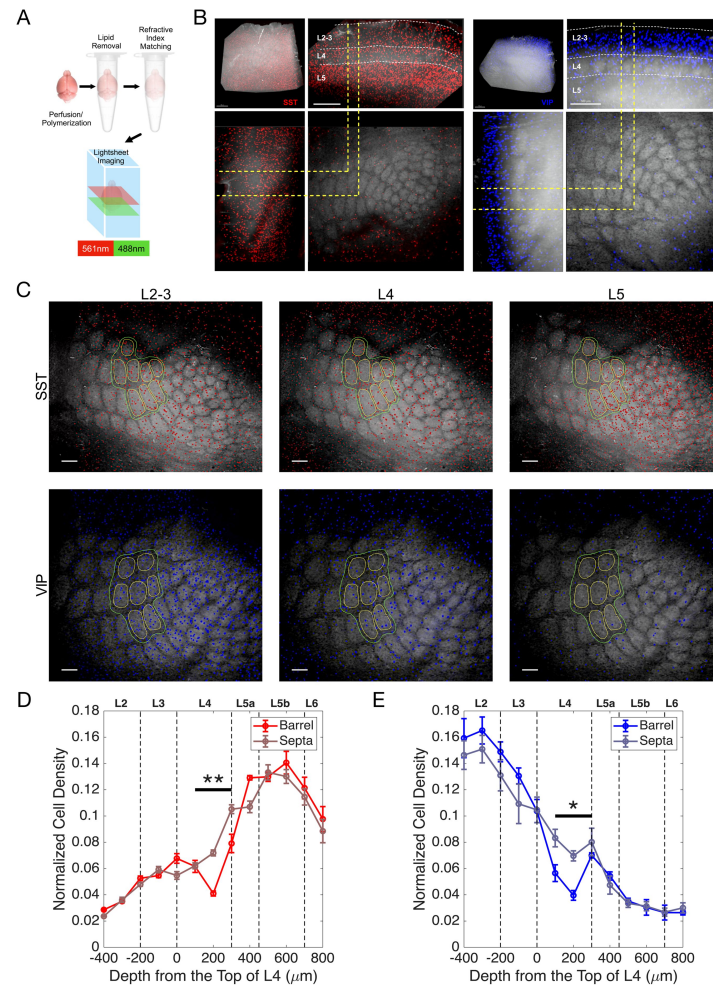


Figure 2.

SST+ and VIP+ neuron densities differ at Barrel and Septa Domains in Layer 4.

(A) A passive CLARITY-based tissue clearing protocol was performed on the collected brains which were subsequently imaged in their entirety using a custom-built light-sheet microscope (mesoSPIM). 561nm wavelength was used to image tdTomato, 488nm wavelength was used to image autofluorescence. (B) 3D reconstruction of cleared SST-Ai14 (left) and VIP-Ai14 (right) barrel cortex. X-Y, X-Z and Y-Z views are presented focusing with a single barrel max projection. The brain was rotated to achieve the orientation that the barrel columns are perpendicular to the X-Y view. The scale bars are set for 500µm. (C) Max projection examples of SST (red) and VIP (blue) neurons in wS1 L2-3, L4 and L5. Quantified barrels (yellow lines) and septa (green lines) are circled out which are perpendicular to X-Y (top-down) view. The scale bar is set for 200µm. (D) SST+ neuron density distribution over laminae for barrel and septal domains (N=4 mice, n=54 Barrels and surrounding septa per barrel). Normalized cell densities plotted as a function of depth. 0µm represents the top of layer 4. SST+ interneuron density is significantly higher in septa ($p=0.0015$, $F=30.64$, repeated-measures ANOVA). (E) VIP+ neuron density distribution over laminae for barrel and septal domains. (N=4 mice, n=53 Barrels and surrounding septa per barrel). Normalized cell densities plotted as a function of depth. 0µm represents the top of layer 4. VIP+ interneuron density is also significantly higher in septa ($p=0.021$, $F=9.63$, repeated-measures ANOVA).

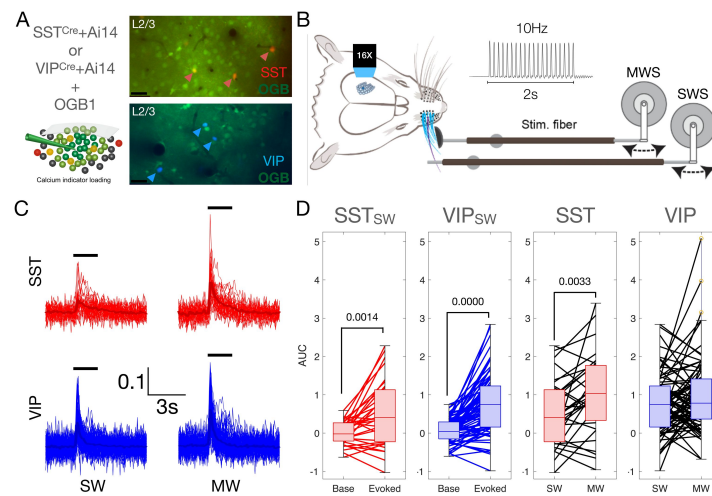


Figure 3.

Single- vs multi-whisker responses of SST+ and VIP+ Interneurons.

(A) Interneurons are labeled with tdTomato using reporter mouse lines and Ca^{2+} imaging has been performed after bulk loading of OGB-1. scale bar 35 μm. (B) Schematic representation of the single- and multi-whisker stimulation protocol upon 10Hz stimulation. (C) $\Delta F/F$ traces of evoked activity of SST (on top) and VIP (on bottom) interneurons ($N = 3$ animals per group, VIP P21+: 138 cells, SST P21+: 51 cells). (D) Area under the curve (AUC) of the first 2s of the $\Delta F/F$ traces of all SST and VIP cells upon single- (SW: 2s-long principal whisker evoked activity) and multi-whisker (MW: 2s-long multi whisker stimulation evoked activity) stimulation (Wilcoxon signed-rank test, Base: 2s-long baseline activity).

domain-assignment strategy as in WT animals (**Figure 1A** and **B**; and **Supp. Figure 1**), we recorded from barrel and septa domains in *Elfn1* KO mice under both single-whisker stimulation (SWS) and multi-whisker stimulation (MWS) conditions. Unsupervised t-SNE analysis revealed that the domain-specific clustering of responses seen in WT (**Figure 1C**) mice was disrupted in *Elfn1* KO mice, with barrel and septa responses intermingling (**Figure 4A**). This convergence of response profiles was further clear in stimulus-aligned stacked images, where the emergent differences between barrels and septa under SWS were largely abolished in the KO (**Figure 4B**). The only remaining distinction was between directly stimulated barrels and neighboring barrels, which persisted in the KO (**Figure 4B**). Under MWS conditions, the temporal divergence between barrel and septa domains observed in WT Layer 4 was diminished in *Elfn1* KO mice (**Figure 4C**). Additionally, the progressive increase in the MW/SW response ratio seen in septal neurons of WT mice was no longer seen in the KO (**Figure 4D**).

These findings suggest that in WT animals, activity spillover from principal barrels is normally constrained by the progressive engagement of SST+ interneurons, driven by *Elfn1*-dependent facilitation at their excitatory synapses. In the absence of *Elfn1*, this local inhibitory mechanism is disrupted, leading to a loss of the distinct temporal response divergence between barrel and septa domains.

Barrel and Septa response decoding identity is lost in *Elfn1* KO mice

Building on our findings that *Elfn1*-dependent facilitation in SST+ interneurons is critical for maintaining barrel-septa response divergence, we sought to quantify how sensory information accumulates and is integrated over time in these domains—and how this process is altered in *Elfn1* KO mice.

To investigate this, we performed an accumulative temporal decoding analysis using neural responses to 20 pulses of either single-whisker stimulation (SWS) or multi-whisker stimulation (MWS) at 10 Hz for 2 seconds. We segmented the response window into three time ranges to identify the temporal segments driving divergence between domains: the full response window (1-95ms), the first part (1-50ms) which includes the first peak, and the late part (51-95ms), which includes the second firing peak (**Figure 5A**). A one-versus-all error-correcting output code (ECOC) classifier, using a GentleBoost ensemble of decision trees (Allwein et al., 2001; Friedman et al., 2000), was trained on firing profiles from WT barrel, septa, and neighboring barrel domains (3-class classification for SWS) or barrel and septa domains (2-class classification for MWS). We conducted 10-fold cross-validation within WT and KO datasets (WT CrossVal and KO CrossVal) and tested a WT-trained classifier on *Elfn1* KO responses (WT Train - KO Test) to evaluate decoding performance across accumulated pulses (see Methods).

In WT mice, decoding accuracy increased progressively with the number of pulses upon SWS (**Figure 5B**, red) reflecting accumulated segregation of sensory information over time. This SWS WT CrossVal trend was more pronounced in the first response window (1-50ms) (**Figure 5C** vs **5D**, red), indicating that domain-specific information accumulates predominantly in first temporal segments in repeated single whisker sampling. Cross-validation within the WT dataset (WT CrossVal) showed higher accuracy compared to the WT-trained classifier tested on KO responses (WT Train - KO Test) and KO CrossVal at this first segment, but not in the second (**Figure 5B-D**). *Elfn1* KO dataset showed higher cross validation performance at the second part of the response profiles (**Figure 5D**) indicating a temporal information delay under the reduced progressive inhibition. While decoding curves started at similar initial levels for SWS data (**Figures 5B-D**, first pulse number), initial MW decoding accuracies were remarkably different (**Figures 5E-G**). In WT mice, while the decoding accuracy increased progressively as in SWS

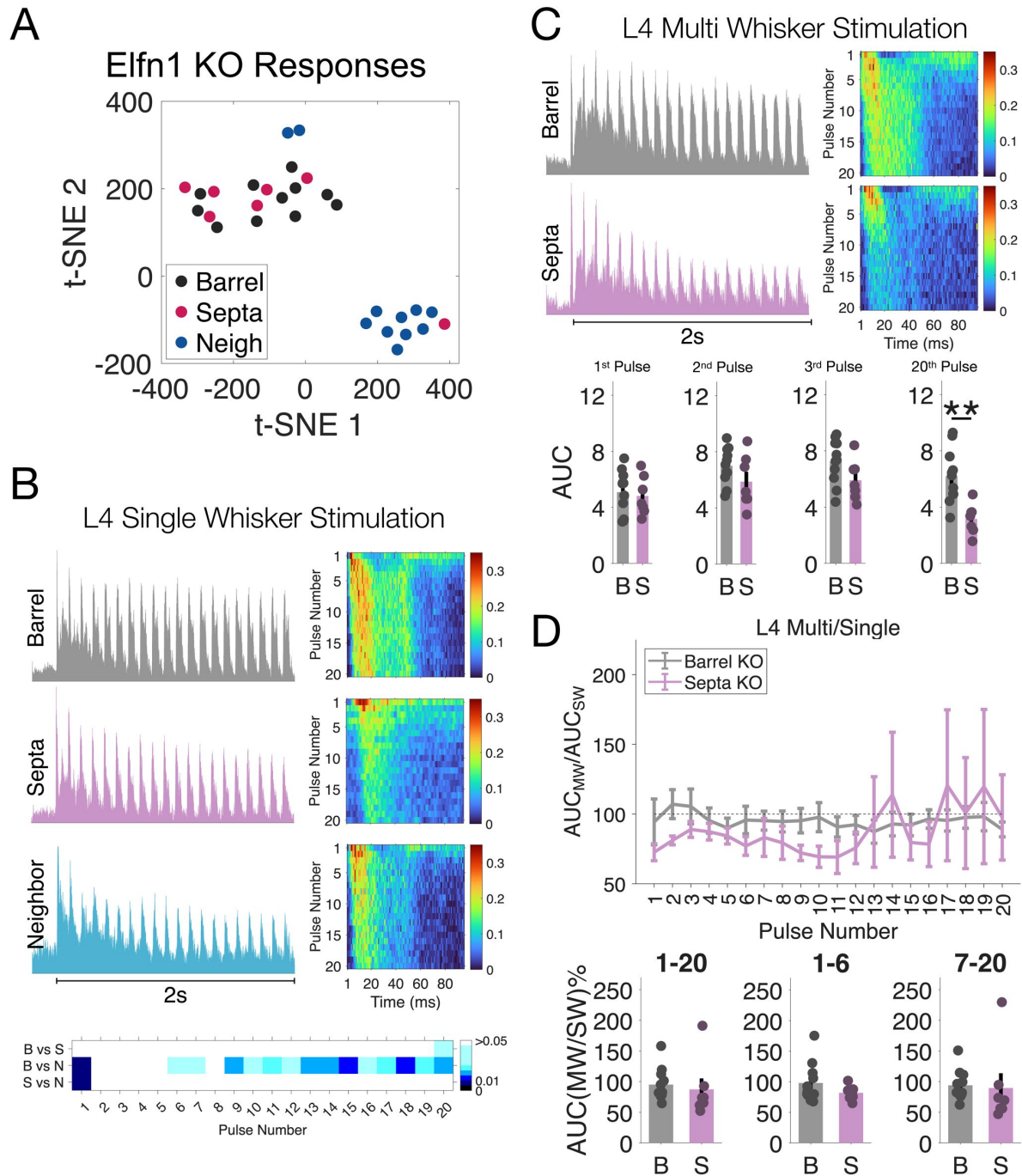


Figure 4.

Loss of Elfn1 abolishes barrel-septa response divergences upon single-whisker stimulation.

(A) t-SNE analysis of the average firing dynamics from three domains. (B) Left: Average multi-unit firing profiles recorded at Layer 4 of barrel (gray), unstimulated neighbor (light blue) and septa (purple) columns (11-barrel columns, 7 septal columns, 11 unstimulated neighbors from N=4 mice) upon 2s-long 10Hz repeated single whisker stimulation. Middle: Matrix representation of the same data by representing each stimulation pulse response on one row. Bottom: Pair-wise statistical comparisons of three conditions for all pulses (Mann-Whitney-U-Test). (C) Same with but response profiles upon MW stimulation. (D) Top: Ratio of MW/SW dynamics for barrel and septa. Bottom: statistical comparison of average ratios of 20 average of all pulses, first six or last 14 pulses, respectively. (p values are color coded in B, stars represents *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$ in C and D.)

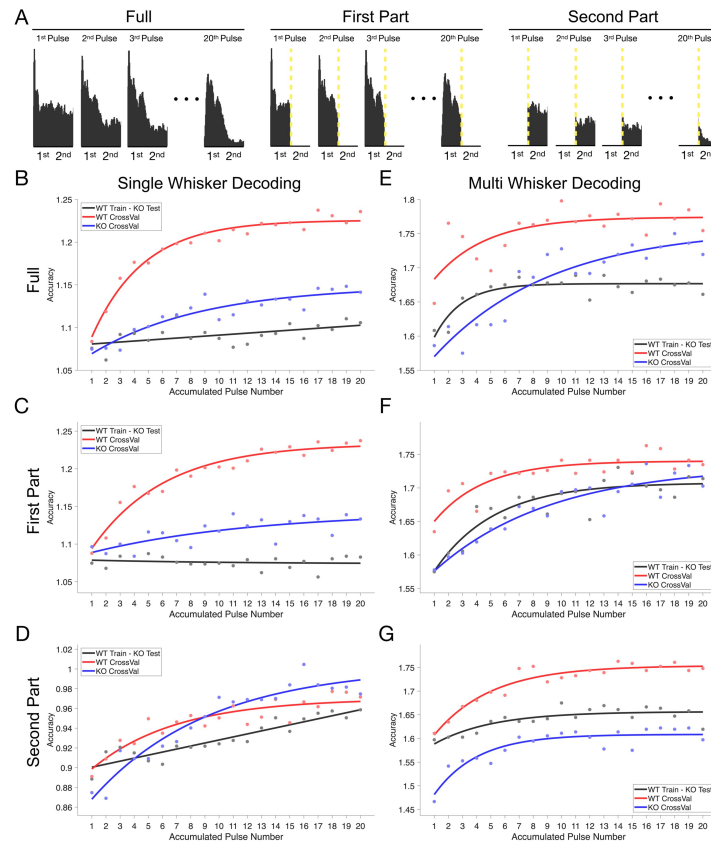


Figure 5.

Accumulative decoding analysis shows an alteration of columnar domain identity in Elnf1 KO animals.

(A) For the decoding analysis, the entire pulse response, only the first 50ms or only the last 45ms has been used. (B) Decoder analysis of the full pulse response profiles in L4 upon SWS. (C) Decoder analysis of the first part of the pulse response (1-50ms) profiles in L4 upon SWS. (D) Decoder analysis of the second part of the pulse response (51-95ms) profiles in L4 upon SWS. (E) Decoder analysis of the full pulse response profiles in L4 upon MWS. (F) Decoder analysis of the first part of the pulse response (1-50ms) profiles in L4 upon MWS. (G) Decoder analysis of the second part of the pulse response (51-95ms) profiles in L4 upon MWS.

(**Figure 5E** [↗](#), red), the difference between accumulated 20th and the starting (1st) accuracies were lower than KO CrossVal results (**Figures 5E-G** [↗](#), red vs blue) which was predominantly due to the first part of the signal (**Figure 5F** [↗](#) vs **5G** [↗](#)).

These results highlight the critical role of *Elfn1* in facilitating the temporal integration of sensory inputs through its effects on SST+ interneurons. In WT mice, *Elfn1*-dependent short-term facilitation enables SST+ interneurons to progressively enhance their inhibitory influence, maintaining the functional segregation between barrel and septa domains. In *Elfn1* KO mice, the loss of this facilitation mechanism disrupts the temporal dynamics necessary for accumulating domain-specific information, leading to a failure in segregating sensory streams. This finding reinforces the importance of local inhibitory circuits, particularly those involving SST+ interneurons, in shaping the temporal processing of sensory information in the barrel cortex.

wS1 Barrel Columns and Septal Domains have different wS2 and M1 Layer-dependent projection preferences

We identified that temporal MW/SW response divergence in septal populations is abolished in *Elfn1* KO mice. Given that barrel and septal compartments differentially process sensory input over time, this raises the question of how these distinct streams of information are relayed to downstream regions and whether they contribute to separate functional pathways. Following up from this observation, we finally aimed to explore if the projection targets of these two wS1 domains are distinct, in a similar manner to what has been revealed for visual cortex domain projections (Meier et al., 2021 [↗](#)). Previously, it has been shown that cortical pyramidal neurons in wS1 exhibit distinct long-range projection patterns and sensory tuning properties depending on their projection target (S2 or M1). S2-projecting pyramidal neurons tend to have narrower receptive fields, are more likely to be tuned to the columnar whisker and carry more precise sensory information about whisker deflections. In contrast, M1-projecting pyramidal neurons are more broadly tuned, integrating information from multiple whiskers, which aligns with their role in sensorimotor integration and behavior (Sato and Svoboda, 2010 [↗](#); Yamashita et al., 2013 [↗](#)). This data could be suggestive of distinct localization of the different projection neurons in the different wS1 domains. By utilizing *in vivo* retrograde AAV injections in two of the main target cortical regions of wS1, we aimed to characterize the projections from wS1 to wS2 and M1.

We injected a retrograde AAV virus expressing GFP in wS2 or M1 in separate adult mice and dissected the brains 4 weeks after the injection (**Figure 6A** [↗](#)). We utilized again the passive CLARITY-based tissue clearing and imaged the whole brain using mesoSPIM. The auto-fluorescence of the barrels allowed for labeling of barrel columns in 3D (**Figure 6B** [↗](#)). With the same strategy as our previous analysis on interneuron population distributions (**Figure 2** [↗](#)), we counted the number of neurons in wS1 barrel and septa domains across the cortical layers (**Figure 6C** [↗](#)) and measured the corresponding domain volumes to estimate projection densities as a proxy for connectivity levels between S1-S2 and S1-M1.

As an internal control and in line with expectations, fewer projections arose from layer 4 barrels to either wS2 or M1 than from other cortical layers (**Figure 6C** [↗](#) and **D** [↗](#)). However, septa cells between mid-L4 and upper-L5B sent more wS2 projections (**Figure 6D** [↗](#), left, turquoise), whereas more neurons in septa from L3 to L4 project to M1, compared to the projection cells in barrels (**Figure 6D** [↗](#), right, magenta). Additionally, a secondary comparative analysis showed that barrels in L3 host significantly more wS2-projecting-neurons (**Figure 6E** [↗](#), left, purple), while L5A septal domains host significantly more M1-projecting neurons (**Figure 6E** [↗](#), right, magenta). This differential targeting suggests that the temporal whisker-evoked processing specialization of barrel and septa domains may shape how distinct streams of sensory information are conveyed to higher-order cortical areas, potentially influencing perceptual processing in wS2 and sensorimotor integration in M1.

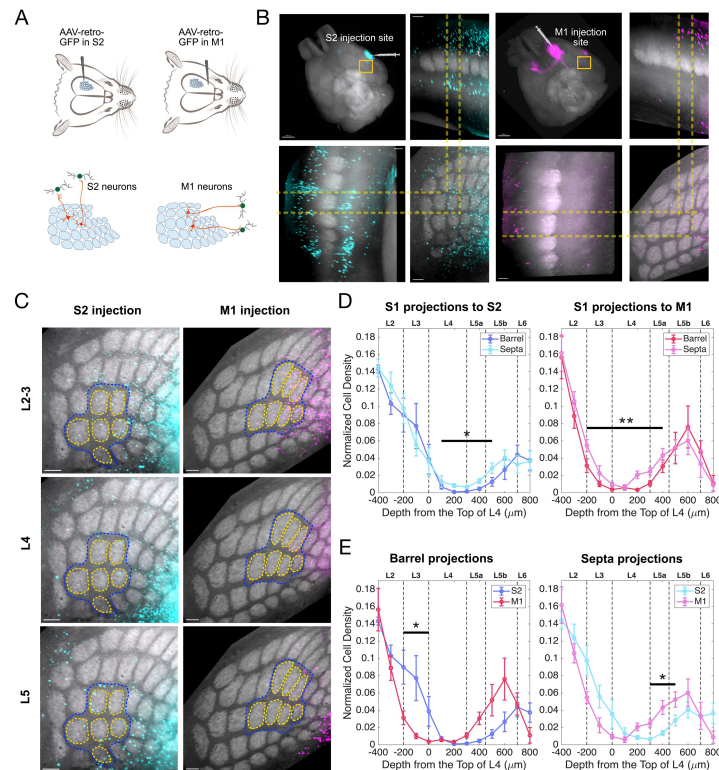


Figure 6.

wS1 barrel and septa columns differentially project to wS2 and M1.

(A) The schematics of the experimental protocol. The retro-AAV-CAG-GFP was injected in either S2, or M1 at P20-30 and the brains were dissected 4 weeks after the virus injection following the tissue clearing and whole brain imaging. (B) 3D reconstruction of the cleared brain with S2(Left panels) or M1(right panels) injection. X-Y, X-Z and Y-Z views are presented focus with a single barrel max projection. The brain was rotated to achieve the orientation that the septa is perpendicular to the X-Y view (top-down). Scale bars in whole brain view are set for 2000um and the scale bars in X-Y, X-Z and Y-Z views are set for 200um. (C) Max projection examples of S2-projected wS1 neurons and M1-projected wS1 neurons in L2-3, L4 and L5 in barrel vs septa. Quantified barrels (yellow lines) and septa (blue lines) are circled out which are perpendicular to X-Y (top-down) view. Scale bars are set for 200um. (D) Left: Normalized cell density profiles of S2 projection neurons at barrel and septa columns. 0μm represents the top of layer 4. L4 and L5a septa columns send significantly more projections to S2 than barrel column (N=4 mice) ($p=0.0139$, $F=11.79$, repeated-measures ANOVA). Right: Normalized cell density profiles of M1 projections neurons at barrel and septa columns. 0μm represents the top of layer 4. L3 and L4 septa columns send significantly more projections to M1 (N=6 mice) ($p=0.0032$, $F=14.82$, repeated measures ANOVA). (E) Left: Normalized cell density profiles of S2 and M1 projection neurons at barrel columns. 0μm represents the top of L4. L3 barrel columns send significantly more projections to S2 (N=4 mice with S2 injections, N=6 mice with M1 injections) ($p=0.0293$, $F=7.02$, repeated-measures ANOVA). Right: Normalized cell density profiles of S2 and M1 projections neurons at septa columns. 0μm represents the top of L4. L5a septal columns send significantly more projections to M1 N=4 mice with S2 injections, N=6 mice with M1 injections) ($p=0.0402$, $F=5.98$, repeated measures ANOVA).

Discussion

In the absence of visual stimuli, humans explore objects through tactile interactions, where the initial touch begins the process of information accumulation, with subsequent touches refining the identification and characterization of the object. Rodents rely on a similar strategy during active whisking, where sequential whisker contacts allow for the progressive refinement of object recognition and spatial mapping (Armstrong-James and Fox, 1987 [↗](#); Kheradpezhohu et al., 2017 [↗](#); Petersen, 2007 [↗](#)). This process involves parallel ascending pathways that transmit peripheral touch sensations to the cortex, with the primary somatosensory cortex mapping whisker spatial patterns directly onto cortical barrels, and in between spaces, septa (Van Der Loos and Woolsey, 1973 [↗](#); Woolsey et al., 1975 [↗](#)).

Our study reveals that the functional segregation between barrel and septa domains in the mouse whisker primary somatosensory cortex (wS1) is not only anatomically defined by distinct thalamocortical projections but is also dynamically shaped by local inhibitory circuits, particularly those involving somatostatin-positive (SST+) interneurons. Through a combination of functional and anatomical techniques, we show that the temporal divergence in spiking activity between barrel and septa domains is mediated by the short-term synaptic facilitation properties of SST+ interneurons and are critically dependent on the synaptic protein Elfn1. Furthermore, using a decoder we show that this temporal divergence may affect the integration of sensory information during repeated whisker stimulation, which may be subsequently routed to differential downstream cortical areas, such as the secondary somatosensory cortex (wS2) and primary motor cortex (M1).

Among the inhibitory neuron classes, SST+ interneurons are known for their high spontaneous activity and powerful regulation of local neural networks through dense feedback connections onto nearby pyramidal neurons (Urban-Ciecko and Barth, 2016 [↗](#)). In contrast, VIP+ interneurons influence network function through a disinhibitory circuit, inhibiting SST+ (and to a lesser extent PV+) interneurons upon activation, which in turn reduces inhibition on pyramidal cells (Karnani et al., 2016 [↗](#); Lee et al., 2013 [↗](#); Pi et al., 2013 [↗](#)). Thus SST+ interneurons may contribute to barrel-septa identity using surround inhibition, while being regulated themselves by VIP+ interneuron activity under certain behaviorally relevant sensory-motor tasks. Our analysis of SST+ cell distribution, shows a higher density in the septa in L4, compared to barrels. Interestingly, layer 4 SST+ interneuron axons remain within layer 4 (Xu et al., 2013 [↗](#)) in the somatosensory cortex, unlike SST+ of layers 2/3 and 5/6 which are Martinotti type and target layer 1. In addition, while Martinotti cells receive facilitating synapses from pyramidal cells and form inhibitory synapses onto dendrites of neighboring pyramidal cells (Silberberg and Markram, 2007 [↗](#)), non-Martinotti SST cells predominantly inhibit fast spiking cells in layer 4 (Ma et al., 2006 [↗](#); Xu et al., 2013 [↗](#)). During naturalistic whisking, SST+ and PV+ cells show divergent activation patterns, with FS cells showing transient excitation due to rapid synaptic depression, while SST+ cells show the opposite (Tan et al., 2008 [↗](#)). It is currently unclear though under which behavioral circumstances SST+ cells of L4 would engage in inhibition of excitatory cells versus inhibition of PV+ cells. Our data suggest that under our experimental conditions, a bottom-up activation paradigm, SST+ cells overall engage in the progressive inhibition of the local L4 circuit, rather than disinhibition. To assess whether temporal divergence in spiking carries meaningful sensory information, we conducted a detailed accumulative temporal decoding analysis. We find that in wild-type mice, decoding accuracy for SWS increased progressively with each additional pulse, particularly in the initial response window (1-50ms), indicating that domain-specific information accumulates primarily in early temporal segments during repeated whisker sampling. This progressive accumulation of decodable information was severely disrupted in Elfn1 KO mice, with cross-validation showing significantly reduced performance compared to wild-type, especially in early response segments. Interestingly, Elfn1 KO mice showed higher cross-validation performance in

the later response segment (51-95ms), suggesting a temporal delay in information processing due to reduced progressive inhibition. Under multi-whisker stimulation, wild-type and knockout mice showed markedly different initial decoding accuracies, with the difference between accumulated and initial accuracies being lower in wild-type than in *Elfn1* KO mice. Our anatomical connectivity mapping would suggest that disruption in temporal information processing in the two domains is crucial for maintaining the functional segregation of parallel processing streams to wS2 and M1.

The differential projection patterns of barrel and septa domains to wS2 and M1 further suggest that the temporal processing capabilities of these domains influence how sensory information is integrated with motor output. For example, the projection of septa domains to M1 may facilitate the rapid translation of sensory information into motor commands, while the projection of barrel domains to wS2 may support more detailed sensory discrimination. The disruption of the informational integrity of these projections in *Elfn1* KO mice could therefore have significant implications for sensorimotor integration and behavior.

It is important to note that while our study focuses on the mouse barrel cortex, there are significant differences between rodent species, such as rats and mice, in terms of cortical organization and whisker system function. Most research on whisker stimulation had traditionally focused on rats (Estebanez et al., 2016 [DOI](#); Petersen, 2007 [DOI](#); Roy et al., 2011 [DOI](#); Tanke et al., 2018 [DOI](#)). The existence of septa in the mouse barrel cortex has been debated, with some suggesting that the distinction between barrel and septal circuits is less pronounced in mice compared to rats (Bureau et al., 2006 [DOI](#); Sato et al., 2007 [DOI](#)). For example, the mouse barrel cortex has fewer neurons per column and thinner septa compared to rats, which may influence the degree of functional segregation between barrel and septa domains. While many animals have whiskers, only a subset of animals have visible cortical barrels. For instance, cats have whiskers and brainstem barrelettes but lack cortical barrels (Nomura et al., 1986 [DOI](#); Rice, 1985 [DOI](#); Rice et al., 1985 [DOI](#)). The presence of barrels in the cortex seems to correlate with volitional whisking behavior, underlining the importance of one-to-one mapping for processing individual whisker information. Our findings show that septa are crucial in separating these parallel whisker streams by modulating the lateral cortico-cortical information flow during repetitive whisker touches. This modulation is significantly influenced by the short-term facilitation properties of synaptic inputs onto SST+ interneurons, facilitated by the *Elfn1* gene, alongside distinct bottom-up inputs to each domain. The loss of temporal segregation in *Elfn1* KO mice could hinder object localization, which relies on multi-whisker integration in septa, while sparing simpler detection tasks mediated by barrels.

In summary, our study demonstrates that there exist a functional segregation of barrel and septa domains in the mouse whisker somatosensory cortex, and that is dynamically shaped by the temporal dynamics of SST+ interneuron activity, imparted by the synaptic protein *Elfn1*. This temporal divergence is essential for the integration of sensory information over repeated stimulation and for the differential projection of information to downstream cortical areas. These findings highlight the importance of local inhibitory circuits in shaping cortical processing and offer new insights into mechanisms underlying sensory perception and behavior.

Methods

Resource availability

Lead Contact

Further information and requests for resources and reagents should be directed to the Lead Contact, T.K. (karayannis@hifo.uzh.ch)

Materials Availability

This study did not generate new unique reagents.

Data and Code Availability

The datasets and analysis routines are available from the first author on reasonable request.

Mice

All the animal experiments followed the guidelines of the Veterinary Office of Switzerland and were approved by the Cantonal Veterinary Office Zurich and the University of Zurich. husbandry with a 12-h reverse dark-light cycle (7 a.m. to 7 p.m. dark) at 24 °C and variable humidity. Adult (5- to 10-week-old) male C57BL6J wide type mice were used for retrograde tracing. Animal lines used in this study are VIP-IRES-Cre ($Vip^{tm1(cre)Zjh/J}$) (Taniguchi et al., 2011 [DOI](#)), SST-IRES-Cre ($Sst^{tm2.1(cre)Zjh/J}$) (Taniguchi et al., 2011 [DOI](#)), Ai14 (B6;129S6-Gt(ROSA)26Sor^{tm14(CAG-tdTomato)Hze/J}) (Madisen et al., 2010 [DOI](#)) and *Elfn1* KO (*Elfn1*^{tm1(KOMP)Vlclg}). Both male and female mice were used in all experiments.

In-Vivo Silicon Probe Recording

We used 4 *Elfn1* KO and 4 WT littermates at the age of P20 to P30 for the multi-electrode silicon probe recordings. Mice were under urethane induced anesthesia (1.5g/1kg) throughout the experiment. A heating pad was used to maintain the mouse's body temperature at 37°C. The depth of anesthesia was checked with breathing speed and paw reflexes throughout the experiment. The skull of the right hemisphere was exposed by removing the skin on top, and a metallic head holder was implanted on the skull with cyanoacrylate glue and dental cement. A 26G needle was used to open a ~2mm x 2mm cranial window which exposed the S1 barrel field (wS1). Extreme care was taken not to cause damage or surface bleeding during surgery.

wS1 neural activities were recorded with an 8-shank-64-channel silicon probe. Each of the 8 shanks has 8 recording sites (100µm apart). The distance between each shank is 200µm (NeuroNexus Technologies, Ann Arbor, MI, USA). The silicon probe was labeled with DiI (1,1'-dioctadecyl-3,3,3',3'-tetramethylindocarbocyanine, Molecular Probes, Eugene, OR, USA) and inserted perpendicularly into the barrel cortex. A silver wire was placed into the cerebellum as a ground electrode. All data were acquired at 20 kHz and stored with MC Rack (RRID:SCR_014955) software (Multi Channel systems). The total duration of multi-electrode recordings varied between 3 to 5 hours. After each experiment, the animal was deeply anesthetized by ketamine (120 mg/kg, ketamine, 50 mg/mL, HamelnPharma, Hameln Germany) and perfused through the aorta with ringer solution. The brain was kept in 4% PFA. Tangential sections (200-µm thick) were prepared for vesicular glutamate transporter 2 (vGlut2, Synaptic Systems, 135404; 1:1000) immunohistochemistry. By combining the DiI and vGlut2 staining, the insertion position of the 8 shank probes were identified in the barrel cortex.

Whisker Stimulation

Either a single principal whisker (SW, single whisker) or multiple whiskers (MW, multi whisker) were stimulated 1mm from the snout in rostral-to caudal direction (about 1mm displacement) using a stainless-steel rod (1 mm diameter) connected to a miniature solenoid actuator. The movement of the tip of the stimulator bar was measured precisely using a laser micrometer (MX series, Metralight, CA, USA) with a 2500 Hz sampling rate. The stimulus takes 26ms to reach the maximal 1 mm whisker displacement, with a total duration of 60ms until it reaches baseline (Yang et al., 2017 [DOI](#)). Both SW and MW stimulation was performed at 10Hz for 2s and each stimulation repeated for 20 times (20 Trials). The inter-trial interval was 30s. In every experiment, multiple whiskers were singly stimulated to increase the probability of finding a proximal barrel-septa and

neighboring barrel triplets. We first performed multi-whisker stimulation and then individual single whisker stimulations were performed at the end. Individual whisker stimulation was performed by attaching a pipet tip to the stimulator, which was then placed in contact with the principal whisker under a dissecting scope. MW stimulation was performed by contacting multiple whiskers with 1cm-wide masking tape attached to the pipet tip.

In vivo two-photon calcium imaging

Neuronal ensembles in superficial layers of the principal whisker barrel field mapped by intrinsic signal imaging were bolus-loaded with the AM ester form of Oregon Green BAPTA-1 by pressure injection (OGB-1; 1 mM solution in Ca^{2+} -free Ringer's solution; 2- min injection at 150–200 μm depth) as described previously (Stosiek et al., 2003). The craniotomy was then filled with agarose (type III-A, 1% in Ringer's solution; Sigma) and covered with an immobilized glass plate. Two-photon Ca^{2+} imaging was performed with a Scientifica HyperScope two-photon laser scanning microscope one hour after bolus loading using a Ti:sapphire laser system at 900 nm excitation (Coherent Chameleon; ~120 femtosecond laser pulses). Two-channel fluorescence images of 256×128 pixels at 11.25 Hz (HyperScope galvo-mode) were collected with a 16x water-immersion objective lens (Nikon, NA 0.8). In all, 3–5 separate spots have been imaged per animal. Per imaging spot, 10 trials of 20s-long evoked activity recorded for each of single- and multi-whisker stimulation paradigms. Data acquisition was controlled by ScanImage (Pologruto et al., 2003).

Analysis of calcium imaging data

Ca^{2+} imaging data is analyzed using custom MATLAB scripts. First, fluorescence image time-series for a given region were concatenated. The concatenated data was then aligned using a Fourier domain cross-correlation based subpixel registration algorithm (Guizar-Sicairos et al., 2008) to correct for translational drift (registered on red tdTomato channel and transferred to OGB-1 channel). Average intensity projections of the imaging data were used as reference images to manually annotate regions of interest (ROIs) corresponding to individual neurons. Neurons with somata partly out-of-focus were not included. Ca^{2+} signals were expressed as the mean pixel value of the relative fluorescence change $\Delta F/F = (F - F_0)/F_0$ in each given ROI. F_0 was calculated as the bottom 5% of the fluorescence trace. Neuropil patches surrounding each neuron is defined by all pixels not assigned to a neuronal soma or astrocyte of the corresponding neuron ROI annotation (Peron et al., 2015) (A disk shaped region around the neuron of interest with excluding any intersecting neighboring neuronal ROI). Neuropil correction is performed as $F_{\text{corrected}} = F_{\text{neuron}} - \alpha * F_{\text{neuropil}}$. α is estimated for each imaging spot separately using the formula $F_{\text{blood_vessel}}/F_{\text{surrounding_neuropil}}$ (Kerlin et al., 2010). For each stimulus, the evoked responses of 10 trials were analyzed and the response magnitude expressed as the mean of the evoked $\Delta F/F$ integral (%s; integral of the first 2s response starting at stimulus onset).

Analysis of Silicon Probe Data

Extracellular silicon probe data were analyzed using a custom-made Matlab script (MATLAB (RRID:SCR_001622), Version: 2024a, MathWorks, MA, USA). The raw data signal was band-pass filtered (0.8-5 kHz), and the multi-unit activity (MUA) was extracted with the threshold of 7.5 times the standard deviation (SD) of baseline (Yang et al., 2017). The current source density (CSD) map was used to identify laminar probe contact location. The earliest CSD sink was identified as layer 4, followed by L2-3 (Reyes-Puerta et al., 2015; Van Der Bourg et al., 2017). MUA was smoothed using a Gaussian kernel (0 mean, 5ms sigma).

Decoder Analysis

Recordings from the principal barrel, adjacent septa, and a neighboring unstimulated barrel were organized into three matrices for wild-type (WT) animals: a $280 \times 95 \times 20$ matrix for the stimulated barrel (14 Barrels, 95ms, 20 pulses), a $180 \times 95 \times 20$ matrix for the septa (9 Septa, 95ms, 20 pulses),

and a $360 \times 95 \times 20$ matrix for the neighboring barrel (18 Neighboring barrels, 95ms, 20 pulses). Each matrix's dimensions correspond to the number of recording channels, time points (95ms epochs), and 20 individual stimulation trials, respectively. Although each trial originally spanned 2000ms (20 pulses at 10Hz), the analysis focused on the first 95ms post-stimulation (95ms instead of 100ms). To examine the impact of different temporal windows on classification performance, three versions of the analysis were performed by varying the time-range parameter, 1 to 95ms for the full epoch, 1 to 50ms for the first part, and 51 to 95ms for the second part.

For each stimulation trial, data was accumulated pulse-by-pulse. In each iteration (ranging from 1 to 20), the corresponding segments from the three matrices were reshaped into two-dimensional feature matrices by concatenating the selected time points across the accumulated pulses. Labels were then assigned to the reshaped data ('B' for the barrel, 'S' for septa, and 'N' for the neighboring barrel).

We employed a classification approach utilizing error-correcting output codes (ECOC) models with a gentle adaptive boosting (GentleBoost) strategy to train classifiers. This method is particularly effective in handling imbalanced data and unequal misclassification costs. Like Logit Boost, each weak learner in the ensemble fitted a regression model to response values, $y_n \in \{-1, +1\}$. The mean-squared error is computed as:

$$N \sum_{n=0}^d d(t)_n (y_n - h_t(x_n))^2$$

where $d(t)_n$ represents observation weights at step t (summing up to 1) and $h_t(x_n)$ denotes predictions from the regression model h_t fitted to response values y_n . As the strength of individual learners diminishes, the weighted mean-squared error converges towards 1.

Three distinct evaluations were performed. WT-to-KO Prediction, the ECOC classifier trained on WT data (compiled from the three regions) was used to predict labels from corresponding recordings in knockout (KO) mice. Within-Group Cross-Validation (WT), A cross-validation procedure was applied to the WT dataset using MATLAB's *crossval* function to assess classifier performance internally. Within-Group Cross-Validation (KO), Similarly, an ECOC classifier was trained and cross-validated on the KO dataset.

For each evaluation, confusion matrices were generated and summary accuracy statistics computed. Accuracy was tracked as a function of the number of accumulated pulses (from 1 to 20). The accuracy value represents the average per-class accuracy. In other words, for each class the accuracy is defined as: $\text{Accuracy}_{\text{class}} = (\text{TP} + \text{TN}) / \text{Total samples}$, where for that class, TP (True Positives) is the number of correctly predicted samples (the corresponding diagonal element of the confusion matrix), and TN (True Negatives) is the sum of all correct predictions for samples not in that class (i.e., all the entries outside the row and column of that class). The function first computes the confusion matrix C where rows represent actual classes and columns represent predicted classes. It then calculates the total number of samples as the sum of all entries in C , and it determines the number of classes. The overall accuracy reported is computed via the formula:

$$\text{Accuracy} = (2 * \text{trace}(C) + 2 * \text{Total}) / ((\text{Number of Classes}) * \text{Total})$$

Here, $\text{trace}(C)$ is the sum of the diagonal elements, which means it is the sum of the true positives for all classes, and Total is the sum of all entries in the confusion matrix. Although this formula may look a bit unusual, it is designed to yield the average of the per-class accuracies. In practice, for each class, one would compute $(\text{TP} + \text{TN}) / \text{Total}$ and then average these values over the classes

that reflects how well the classifier performed across all classes. To characterize the evolution of classification accuracy with increasing pulse number, non-linear exponential functions of the form:

$$f(x) = a - b * \exp(-cx)$$

were fitted to the accuracy data using a non-linear least squares approach (implemented via the *fminsp* function). Separate fits were obtained for the WT-to-KO prediction, WT cross-validation, and KO cross-validation accuracy curves. The fitted curves and raw accuracy data were then plotted. This integrated analysis pipeline allowed for systematic comparison of classification performance across spatial locations and conditions and an assessment of the temporal dynamics underlying the neural responses to single-whisker stimulation. The same strategy has been used for multi-whisker decoding but between only Barrel and Septa.

Retrograde tracing

Mice aged from P10-37 were anesthetized by isoflurane and a small craniotomy was made above the injection site (M1 or S2). 200nl retrograde AAV virus, ssAAV-retro/2-CAG-EGFP-WPRE-SV40p(A) (physical titer: 5×10^{12} vg/ml) was injected 200-500µm deep in the M1 or S2. Virus was purchased from Zurich Viral Vector Facility (Cat#: V24-retro).

Tissue Processing for Passive Clearing and Imaging

The method used for hydrogel-based tissue clearing is explained in detail elsewhere (Chung and Deisseroth, 2013 [DOI](#)). Briefly, Mice age P29-70 (4 weeks after retrograde AAV virus injection) or SST-Ai14 and VIP-Ai14 mice age from P14-77 were transcardially perfused using 1x PBS and Hydrogel solution (1% PFA, 4% Acrylamide, 0.05% Bis). The collected brains were post-fixed for 48 h in a Hydrogel solution (1% PFA, 4% Acrylamide, 0.05% Bis). Afterward, the Hydrogel polymerization was induced at 37 °C. Following the polymerization, the brains were immersed in 40 mL of 8% SDS and kept shaking at room temperature until the tissue was cleared sufficiently (15-90 days depending on the animal's age). Finally, after 2–4 washes in PBS, the brains were put into a self-made refractive index matching solution (RIMS). They were left to equilibrate in 5 mL of RIMS for at least 4 days at RT (Room Temperature) before being imaged. After clearing, brains were attached to a small weight and loaded into a quartz cuvette, then submerged in RIMS, and imaged using a home-built mesoscale selective plane illumination microscope (mesoSPIM, Voigt et al., 2019 [DOI](#)). The microscope consists of a dual-sided excitation path using a fiber-coupled multiline laser combiner (405, 488, 515, 561, 594, 647 nm, Omicron SOLE-6) and a detection path comprising an Olympus MVX-10 zoom macroscope with a 1× objective (Olympus MVPLAPO 1x), a filter wheel (Ludl 96A350), and a scientific CMOS (sCMOS) camera (Hamamatsu Orca Flash 4.0 V3). For imaging tdTomato and eGFP a 594 nm excitation with a 594 long-pass filter (594 LP Edge Basic, AHF) and 488 nm & 520/35 (BrightLine HC, AHF) were used, respectively. The excitation paths also contain galvo scanners (GCM-2280-1500, Citizen Chiba) for light-sheet generation and reduction of streaking artifacts due to absorption of the light-sheet. In addition, the beam waist is scanning using electrically tunable lenses (ETL, Optotune EL-16-40-5D-TC-L) synchronized with the rolling shutter of the sCMOS camera. This axially scanned light-sheet mode (ASLM) leads to a uniform axial resolution across the field-of-view of 5–10 µm (depending on zoom and wavelength). The magnification 0.8X (pixel size: 8.23 µm) was used for the view of the whole brain and 2X (pixel size: 3.26 µm) was used for the view of barrel cortex. Further technical details of the mesoSPIM are described elsewhere (Vladimirov et al., 2023 [DOI](#); Voigt et al., 2019 [DOI](#)).

Analysis of cleared brain imaging data

Barrel cortex was imaged in 3D with 2X objective. The images were processed through Fiji and Imaris to visualize the 3D structure of the barrel cortex. XY-, YZ-, XZ-projections were visualized in Imaris, which allows us to adjust the correct angle for each barrel column to be vertical against

XY-plane. This method allowed us identify barrel and septa in layers 2-3 and 5 and layer 4. In addition, the depth information is also precisely acquired for each barrel column. SST+ or VIP+ neuron numbers, S2-projected and M1-projected S1 neuron numbers were counted in icy. Max projection for each 50 μm s along barrel columns was acquired and neurons on each max projection image were counted automatically by icy (RRID:SCR_010587) with the same threshold for all the brains. Each barrel was drawn according to the barrel map from each brain by max projection. The area of septa was calculated by subtracting the whole selected area from all the selected barrel areas.

t-SNE Analysis

To visualize the segregation of neural responses across the different cortical domains, we applied t-distributed Stochastic Neighbor Embedding (t-SNE) to the averaged datasets. t-SNE was used to reduce the high-dimensional data into a 2D space for visualization. The parameters for t-SNE were set to use the *barnesht* algorithm for efficiency, with a perplexity of 9 and exaggeration of 3 to enhance cluster separation. The resulting t-SNE coordinates were plotted using MATLAB's *gscatter* function to distinguish between the different domains (Barrel, Septa, Neighboring). This analysis allowed us to visually inspect the clustering of neural responses from different cortical domains.

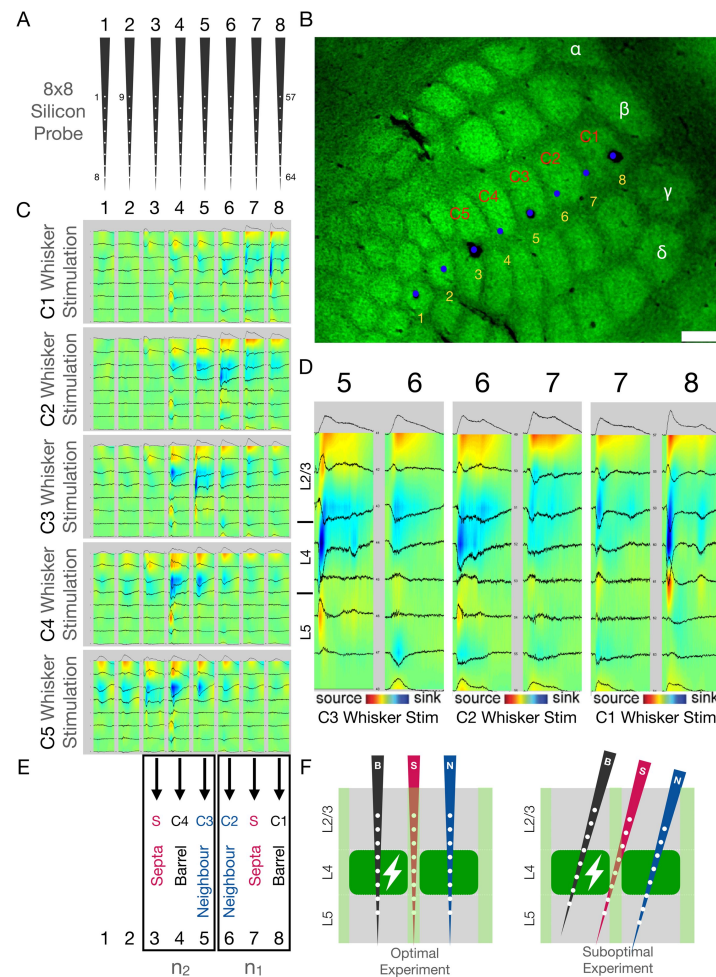
Statistical Analysis of the Multi-Unit Activity

To evaluate the statistical significance of differences in multi-unit activity (MUA) responses between barrel (B), septa (S), and neighboring barrel (N) domains across 20 pulses of whisker stimulation, we performed a pulse-by-pulse comparison using the non-parametric Mann-Whitney U test (also known as the Wilcoxon rank-sum test). This analysis was conducted on the averaged MUA data recorded from Layer 4 of the whisker somatosensory cortex (wS1) in wild-type (WT) and *Elfn1* knockout (KO) mice during single-whisker stimulation (SWS) and multi-whisker stimulation (MWS) paradigms, as detailed above. For each of the 20 stimulation pulses (1 to 20), the MUA data for each domain were reshaped and averaged across trials. The Mann-Whitney U test was then applied pairwise to assess differences between domains: barrel vs. septa (B vs S), barrel vs. neighboring barrel (B vs N), and septa vs. neighboring barrel (S vs N). The resulting p-values were stored in a 3×20 matrix, where rows correspond to the pairwise comparisons and columns represent the pulse number. To visualize the statistical outcomes, a heatmap was generated using MATLAB (Version 2024a, MathWorks, MA, USA). A custom discrete colormap was created by adapting the hot colormap, which originally provides a gradient of 10 colors ranging from black (low values) to white (high values). We selected specific indices from this gradient (10, 8, 6, 4, 2, 1) to define a 10-level discrete colormap, assigning the darkest shade (black) to p-values ≤ 0.05 (indicating statistical significance) and progressively lighter shades to higher p-values, with the lightest shade (white) representing p-values > 0.05 . The colormap was inverted to ensure that significant p-values (≤ 0.05) appeared as light shades against a dark background, enhancing visual contrast. The heatmap was plotted with p-values scaled between 0 and 0.05, using the custom colormap.

Statistical Analysis

Data are represented as mean $\pm$ SEM unless stated otherwise. Statistical comparisons have been done using a one-tailed Mann-Whitney-U test for non-paired data and one-tailed Wilcoxon signed-rank test for paired comparisons (the statistical significance threshold was set to $p < 0.05$; in the Figures, different degrees of evidence against the null hypothesis are indicated by asterisks ($p < 0.05$: *; $p < 0.01$: **; $p < 0.001$: ***). Cell densities in **Figures 2** and **6** were compared using repeated measures ANOVA. All tests were conducted using custom codes in Matlab.

Supplementary Figures



Supplementary Figure 1.

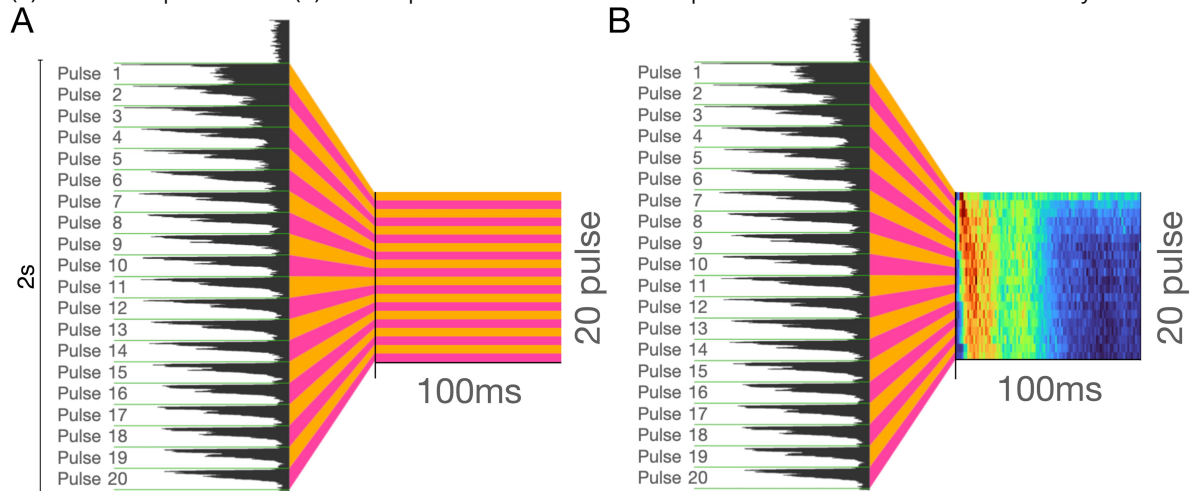
An example of assignment of barrel, septa and Neighbor electrodes using histology and current source density (CSD) analysis.

(A) Shank electrode map of 8x8 silicon probes. (B) Probe locations were labeled (DiI) after each experiment using histology (vGlut2 staining). Matching shank locations are indicated using yellow numbers. Blue dots represent the insertion sites. The red and white text depicts corresponding barrel names (C) All principal whiskers are stimulated one-by-one. C1 to C5 barrel responses upon single whisker stimulation are shown from top to bottom. (D) A closer look on C1, C2 and C3 whisker stimulations and corresponding CSD at shanks 5,6,7 and 8. (E), Barrel (B), adjacent septa (S) and the adjacent neighboring barrel (N) were assigned according to the probe insertion sites and L4 barrel autofluorescence in combination with CSD responses. For example, while shanks 6 and 8 show strong response upon single whisker stimulation (and hence assigned as barrels), shank 7 does not show the expected response (hence assigned as septa). Only the triplets of electrodes were used in the analysis in which principal barrel (B), adjacent septa (S) and the adjacent neighboring barrel (N) were captured by the probe insertion sites for any experiment (Left). 14-barrel columns, 9 septal columns, 18 unstimulated neighboring barrels from N=4 mice. (F) An ideal schematic representation (on the left). Realistically, we cannot always be sure if an entire shank is consistently going through a barrel or septal column due to the curvature of the cortex (on the right) for above and below of layer4.

Supplementary Figure 2.

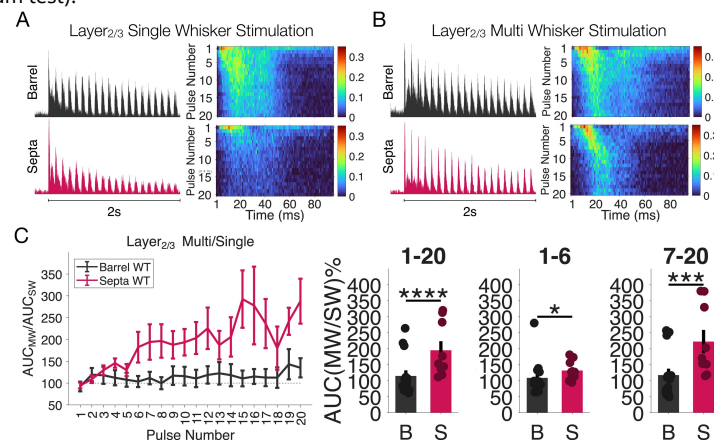
Representation of the conversion of the two-second-long single trial recordings into 20×100ms matrices.

(A) Schematic representation. (B) An example data. The last 5ms of each pulse have been discarded in further analysis.



Supplementary Figure 3.

(A) Left: Average multi-unit firing profiles recorded at Layer 2/3 of barrel (black), unstimulated neighbor (blue) and septa (pink) columns upon 2s-long 10Hz repeated single whisker stimulation. Right: Another representation of the same data but converted into an image by representing each stimulation pulse response on one row. (B) Left: Average multi-unit firing profiles recorded at Layer 2/3 of barrel (black), unstimulated neighbor (blue) and septa (pink) columns upon 2s-long 10Hz repeated multi whisker stimulation. Right: Another representation of the same data but converted into an image by representing each stimulation pulse response on one row. (C) Left: Temporal ratio dynamics (average AUC MW responses divided by average AUC SW responses) for barrel and septa for layer2/3. Right: statistical comparison of average AUC ratios of all pulses (1-20), first six (1-6) or last 14 pulses (7-20), respectively (stars represents *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$ in F and G. Wilcoxon rank sum test).



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Author Contributions

A.Ö.A. curated the data, analyzed in-vivo silicon probe recordings and the two-photon calcium imaging data, performed the decoder analysis and wrote the manuscript. T.J.S. conceptualized the study, maintained *Elfn1* KO line, performed earlier electrophysiological characterizations and contributed to the drafting of the manuscript, J.W.Y. performed silicon probe recordings, AvdB collected two-photon calcium imaging data, L.C. performed light-sheet data analysis, R.K. performed tissue clearing and light-sheet image acquisition, T.K. conceptualized and supervised the study and wrote the manuscript.

References

- Ahissar E, Sosnik R, Bagdasarian K, Haidarliu S (2001) **Temporal frequency of whisker movement. II. Laminar organization of cortical representations** *J Neurophysiol* **86** <https://doi.org/10.1152/jn.2001.86.1.354> | Google Scholar
- Alloway KD (2008) **Information processing streams in rodent barrel cortex: The differential functions of barrel and septal circuits** *Cerebral Cortex* <https://doi.org/10.1093/cercor/bhm138> | Google Scholar
- Allwein EL, Schapire RE, Singer Y (2001) **Reducing multiclass to binary: A unifying approach for margin classifiers** *Journal of Machine Learning Research* **1** | Google Scholar
- Almásí Z, Dávid C, Witte M, Staiger JF (2019) **Distribution patterns of three molecularly defined classes of gabaergic neurons across columnar compartments in mouse barrel cortex** *Front Neuroanat* **13** <https://doi.org/10.3389/fnana.2019.00045> | Google Scholar
- Armstrong-James M, Fox K (1987) **Spatiotemporal convergence and divergence in the rat S1 “Barrel” cortex** *Journal of Comparative Neurology* **263** <https://doi.org/10.1002/cne.902630209> | Google Scholar
- Audette NJ, Urban-Ciecko J, Matsushita M, Barth AL (2018) **POm Thalamocortical Input Drives Layer-Specific Microcircuits in Somatosensory Cortex** *Cereb Cortex* **28** <https://doi.org/10.1093/cercor/bhx044> | Google Scholar
- Brecht M, Sakmann B (2002) **Dynamic representation of whisker deflection by synaptic potentials in spiny stellate and pyramidal cells in the barrels and septa of layer 4 rat somatosensory cortex** *Journal of Physiology* <https://doi.org/10.1113/jphysiol.2002.018465> | Google Scholar
- Bureau I, Von Paul F Saint, Svoboda K. (2006) **Interdigitated paralemniscal and lemniscal pathways in the mouse barrel cortex** *PLoS Biol* **4** <https://doi.org/10.1371/journal.pbio.0040382> | Google Scholar
- Cai L, Yang JW, Wang CF, Chou SJ, Luhmann HJ, Karayannis T (2022) **Identification of a Developmental Switch in Information Transfer between Whisker S1 and S2 Cortex in Mice** *Journal of Neuroscience* **42** <https://doi.org/10.1523/JNEUROSCI.2246-21.2022> | Google Scholar
- Carvell GE, Simons DJ (1990) **Biometric analyses of vibrissal tactile discrimination in the rat** *Journal of Neuroscience* **10** <https://doi.org/10.1523/jneurosci.10-08-02638.1990> | Google Scholar
- Chakrabarti S, Alloway KD (2006) **Differential origin of projections from SI barrel cortex to the whisker representations in SII and MI** *Journal of Comparative Neurology* **498** <https://doi.org/10.1002/cne.21052> | Google Scholar

- Chmielowska J, Carvell GE, Simons DJ (1989) **Spatial organization of thalamocortical and corticothalamic projection systems in the rat SmI barrel cortex** *Journal of Comparative Neurology* **285** <https://doi.org/10.1002/cne.902850304> | Google Scholar
- Chung K, Deisseroth K (2013) **CLARITY for mapping the nervous system** *Nat Methods* **10** <https://doi.org/10.1038/nmeth.2481> | Google Scholar
- Douglas RJ, Martin KAC (2004) **Neuronal circuits of the neocortex** *Annu Rev Neurosci* <https://doi.org/10.1146/annurev.neuro.27.070203.144152> | Google Scholar
- Estebanez L, Bertherat J, Shulz DE, Bourdieu L, Léger JF (2016) **A radial map of multi-whisker correlation selectivity in the rat barrel cortex** *Nat Commun* **7** <https://doi.org/10.1038/ncomms13528> | Google Scholar
- Friedman J, Hastie T, Tibshirani R (2000) **Additive logistic regression: A statistical view of boosting** *Ann Stat* <https://doi.org/10.1214/aos/1016218223> | Google Scholar
- Grant RA, Mitchinson B, Prescott TJ (2012) **The development of whisker control in rats in relation to locomotion** *Dev Psychobiol* **54** <https://doi.org/10.1002/dev.20591> | Google Scholar
- Guizar-Sicairos M, Thurman ST, Fienup JR (2008) **Efficient subpixel image registration algorithms** *Opt Lett* **33** <https://doi.org/10.1364/ol.33.000156> | Google Scholar
- Karnani MMM, Jackson J, Ayzenshtat I, Tucciarone J, Manoocheri K, Snider WGG, Yuste R (2016) **Cooperative Subnetworks of Molecularly Similar Interneurons in Mouse Neocortex** *Neuron* **90** <https://doi.org/10.1016/j.neuron.2016.02.037> | Google Scholar
- Kastli R, Vighagen R, van der Bourg A, Argunsah AO, Iqbal A, Voigt FF, Kirschenbaum D, Aguzzi A, Helmchen F, Karayannis T. (2020) **Developmental divergence of sensory stimulus representation in cortical interneurons** *Nat Commun* **11** <https://doi.org/10.1038/s41467-020-19427-z> | Google Scholar
- Kerlin AM, Andermann ML, Berezovskii VK, Reid RC (2010) **Broadly Tuned Response Properties of Diverse Inhibitory Neuron Subtypes in Mouse Visual Cortex** *Neuron* **67** <https://doi.org/10.1016/j.neuron.2010.08.002> | Google Scholar
- Kheradpezhoh E, Adibi M, Arabzadeh E (2017) **Response dynamics of rat barrel cortex neurons to repeated sensory stimulation** *Sci Rep* **7** <https://doi.org/10.1038/s41598-017-11477-6> | Google Scholar
- Kim U, Ebner FF (1999) **Barrels and septa: Separate circuits in rat barrel field cortex** *Journal of Comparative Neurology* **408** [https://doi.org/10.1002/\(SICI\)1096-9861\(19990614\)408:4<489::AID-CNE4>3.0.CO;2-E](https://doi.org/10.1002/(SICI)1096-9861(19990614)408:4<489::AID-CNE4>3.0.CO;2-E) | Google Scholar
- Koralek KA, Jensen KF, Killackey HP (1988) **Evidence for two complementary patterns of thalamic input to the rat somatosensory cortex** *Brain Res* **463** [https://doi.org/10.1016/0006-8993\(88\)90408-8](https://doi.org/10.1016/0006-8993(88)90408-8) | Google Scholar
- Lee S, Kruglikov I, Huang ZJ, Fishell G, Rudy B (2013) **A disinhibitory circuit mediates motor integration in the somatosensory cortex** *Nat Neurosci* **16** <https://doi.org/10.1038/nn.3544> | Google Scholar
- Ma Y, Hu H, Berrebi AS, Mathers PH, Agmon A (2006) **Distinct subtypes of somatostatin-containing neocortical interneurons revealed in transgenic mice** *Journal of Neuroscience*

26 <https://doi.org/10.1523/JNEUROSCI.0661-06.2006> | Google Scholar

Madisen L, Zwingman TA, Sunkin SM, Oh SW, Zariwala HA, Gu H, Ng LL, Palmiter RD, Hawrylycz MJ, Jones AR, Lein ES, Zeng H (2010) **A robust and high-throughput Cre reporting and characterization system for the whole mouse brain** *Nat Neurosci* **13**:133–140 <https://doi.org/10.1038/nn.2467> | Google Scholar

Meier AM, Wang Q, Ji W, Ganachaud J, Burkhalter A (2021) **Modular network between postrhinal visual cortex, amygdala, and entorhinal cortex** *Journal of Neuroscience* **41** <https://doi.org/10.1523/JNEUROSCI.2185-20.2021> | Google Scholar

Melzer P, Champney GC, Maguire MJ, Ebner FF (2006a) **Rate code and temporal code for frequency of whisker stimulation in rat primary and secondary somatic sensory cortex** *Exp Brain Res* **172**:370–386 <https://doi.org/10.1007/s00221-005-0334-1> | Google Scholar

Melzer P, Sachdev RNS, Jenkinson N, Ebner FF (2006b) **Stimulus frequency processing in awake rat barrel cortex** *Journal of Neuroscience* **26** <https://doi.org/10.1523/JNEUROSCI.2620-06.2006> | Google Scholar

Nomura S, Itoh K, Sugimoto T, Yasui Y, Kamiya H, Mizuno N (1986) **Mystacial vibrissae representation within the trigeminal sensory nuclei of the cat** *Journal of Comparative Neurology* **253** <https://doi.org/10.1002/cne.902530110> | Google Scholar

O'Connor DH, Hires SA, Guo Z V., Li N, Yu J, Sun QQ, Huber D, Svoboda K (2013) **Neural coding during active somatosensation revealed using illusory touch** *Nat Neurosci* **16** <https://doi.org/10.1038/nn.3419> | Google Scholar

Pammer L, O'Connor DH, Hires SA, Clack NG, Huber D, Myers EW, Svoboda K (2013) **The mechanical variables underlying object localization along the axis of the whisker** *Journal of Neuroscience* **33** <https://doi.org/10.1523/JNEUROSCI.4316-12.2013> | Google Scholar

Peron S, Chen TW, Svoboda K (2015) **Comprehensive imaging of cortical networks** *Curr Opin Neurobiol* <https://doi.org/10.1016/j.conb.2015.03.016> | Google Scholar

Petersen CCH (2007) **The functional organization of the barrel cortex** *Neuron* <https://doi.org/10.1016/j.neuron.2007.09.017> | Google Scholar

Pi HJ, Hangya B, Kvitsiani D, Sanders JI, Huang ZJ, Kepecs A (2013) **Cortical interneurons that specialize in disinhibitory control** *Nature* **503** <https://doi.org/10.1038/nature12676> | Google Scholar

Pologruto TA, Sabatini BL, Svoboda K (2003) **ScanImage: Flexible software for operating laser scanning microscopes** *Biomed Eng Online* **2** <https://doi.org/10.1186/1475-925X-2-13> | Google Scholar

Reyes-Puerta V, Sun JJ, Kim S, Kilb W, Luhmann HJ (2015) **Laminar and Columnar Structure of Sensory-Evoked Multineuronal Spike Sequences in Adult Rat Barrel Cortex in Vivo** *Cerebral Cortex* **25** <https://doi.org/10.1093/cercor/bhu007> | Google Scholar

Rice FL (1985) **An attempt to find vibrissa-related barrels in the primary somatosensory cortex of the cat** *Neurosci Lett* **53** [https://doi.org/10.1016/0304-3940\(85\)90180-6](https://doi.org/10.1016/0304-3940(85)90180-6) | Google Scholar

- Rice FL, Gomez C, Barstow C, Burnet A, Sands P (1985) **AComparative analysis of the development of the primary somatosensory cortex: Interspecies similarities during barrel and laminar development** *Journal of Comparative Neurology* **236** <https://doi.org/10.1002/cne.902360405> | Google Scholar
- Roy NC, Bessaih T, Contreras D (2011) **Comprehensive mapping of whisker-evoked responses reveals broad, sharply tuned thalamocortical input to layer 4 of barrel cortex** *J Neurophysiol* **105** <https://doi.org/10.1152/jn.00939.2010> | Google Scholar
- Sato TR, Gray NW, Mainen ZF, Svoboda K (2007) **The functional microarchitecture of the mouse barrel cortex** *PLoS Biol* **5** <https://doi.org/10.1371/journal.pbio.0050189> | Google Scholar
- Sato TR, Svoboda K (2010) **The functional properties of barrel cortex neurons projecting to the primary motor cortex** *Journal of Neuroscience* **30** <https://doi.org/10.1523/JNEUROSCI.3774-09.2010> | Google Scholar
- Shepherd GMG, Svoboda K (2005) **Laminar and columnar organization of ascending excitatory projections to layer 2/3 pyramidal neurons in rat barrel cortex** *Journal of Neuroscience* **25** <https://doi.org/10.1523/JNEUROSCI.1173-05.2005> | Google Scholar
- Silberberg G, Markram H (2007) **Disynaptic Inhibition between Neocortical Pyramidal Cells Mediated by Martinotti Cells** *Neuron* **53** <https://doi.org/10.1016/j.neuron.2007.02.012> | Google Scholar
- Sosnik R, Haidarliu S, Ahissar E (2001) **Temporal frequency of whisker movement. I. Representations in brain stem and thalamus** *J Neurophysiol* **86** <https://doi.org/10.1152/jn.2001.86.1.339> | Google Scholar
- Stachniak TJ, Kastli R, Hanley O, Argunsah AÖ, Van Der Valk EGT, Kanatouris G, Karayannis T. (2021) **Postmitotic prox1 expression controls the final specification of cortical VIP interneuron subtypes** *Journal of Neuroscience* **41** <https://doi.org/10.1523/JNEUROSCI.1021-21.2021> | Google Scholar
- Stachniak TJ, Sylwestrak EL, Scheiffele P, Hall BJ, Ghosh A (2019) **Elfn1-induced constitutive activation of mglur7 determines frequency-dependent recruitment of somatostatin interneurons** *Journal of Neuroscience* **39** <https://doi.org/10.1523/JNEUROSCI.2276-18.2019> | Google Scholar
- Staiger JF, Petersen CCH (2021) **Neuronal circuits in barrel cortex for whisker sensory perception** *Physiol Rev* **101** <https://doi.org/10.1152/physrev.00019.2019> | Google Scholar
- Stosiek C, Garaschuk O, Holthoff K, Konnerth A (2003) **In vivo two-photon calcium imaging of neuronal networks** *Proc Natl Acad Sci U S A* **100** <https://doi.org/10.1073/pnas.1232232100> | Google Scholar
- Tan Z, Hu H, Huang ZJ, Agmon A (2008) **Robust but delayed thalamocortical activation of dendritic-targeting inhibitory interneurons** *Proc Natl Acad Sci U S A* **105** <https://doi.org/10.1073/pnas.0710628105> | Google Scholar
- Taniguchi H, He M, Wu P, Kim S, Paik R, Sugino K, Kvitsani D, Fu Y, Lu J, Lin Y, Miyoshi G, Shima Y, Fishell G, Nelson SB, Huang ZJ (2011) **A Resource of Cre Driver Lines for Genetic Targeting of GABAergic Neurons in Cerebral Cortex** *Neuron* **71**:995–1013 <https://doi.org/10.1016/j.neuron.2011.07.026> | Google Scholar

- Tanke N, Borst JGG, Houweling AR (2018) **Single-cell stimulation in barrel cortex influences psychophysical detection performance** *Journal of Neuroscience* **38** <https://doi.org/10.1523/JNEUROSCI.2155-17.2018> | [Google Scholar](#)
- Tomioka NH, Yasuda H, Miyamoto H, Hatayama M, Morimura N, Matsumoto Y, Suzuki T, Odagawa M, Odaka YS, Iwayama Y, Won Um J, Ko J, Inoue Y, Kaneko S, Hirose S, Yamada K, Yoshikawa T, Yamakawa K, Aruga J (2014) **Elfn1 recruits presynaptic mGluR7 in trans and its loss results in seizures** *Nat Commun* **5** <https://doi.org/10.1038/ncomms5501> | [Google Scholar](#)
- Urban-Ciecko J, Barth AL (2016) **Somatostatin-expressing neurons in cortical networks** *Nat Rev Neurosci* **17** <https://doi.org/10.1038/nrn.2016.53> | [Google Scholar](#)
- Van Der Bourg A, Yang JW, Reyes-Puerta V, Laurenczy B, Wieckhorst M, Stüttgen MC, Luhmann HJ, Helmchen F. (2017) **Layer-Specific Refinement of Sensory Coding in Developing Mouse Barrel Cortex** *Cerebral Cortex* **27** <https://doi.org/10.1093/cercor/bhw280> | [Google Scholar](#)
- Van Der Loos H, Woolsey TA. (1973) **Somatosensory cortex: Structural alterations following early injury to sense organs** *Science (1979)* **179** <https://doi.org/10.1126/science.179.4071.395> | [Google Scholar](#)
- Vladimirov N, Voigt F, Naert T, Araujo GR, Cai R, Reuss AM, Zhao S, Schmid P, Hildebrand S, Schaettin M, Groos D, Mateos JM, Bethge P, Yamamoto T, Aerne V, Roebroek A, Ertürk A, Aguzzi A, Ziegler U, Stoekli ET, Baudis L, Lienkamp SS, Helmchen F. (2023) **The Benchtop mesoSPIM: a next-generation open-source light-sheet microscope for large cleared samples** *bioRxiv* [Google Scholar](#)
- Voigt FF, Kirschenbaum D, Platonova E, Pagès S, Campbell RAA, Kastli R, Schaettin M, Egolf L, van der Bourg A, Bethge P, Haenraets K, Frézel N, Topilko T, Perin P, Hillier D, Hildebrand S, Schueth A, Roebroek A, Roska B, Stoekli ET, Pizzala R, Renier N, Zeilhofer HU, Karayannis T, Ziegler U, Batti L, Holtmaat A, Lüscher C, Aguzzi A, Helmchen F. (2019) **The mesoSPIM initiative: open-source light-sheet microscopes for imaging cleared tissue** *Nat Methods* **16** <https://doi.org/10.1038/s41592-019-0554-0> | [Google Scholar](#)
- Wang HC, LeMessurier AM, Feldman DE (2022) **Tuning instability of non-columnar neurons in the salt-and-pepper whisker map in somatosensory cortex** *Nat Commun* **13** <https://doi.org/10.1038/s41467-022-34261-1> | [Google Scholar](#)
- Woolsey TA, Welker C, Schwartz RH (1975) **Comparative anatomical studies of the Sml face cortex with special reference to the occurrence of “barrels” in layer IV** *Journal of Comparative Neurology* **164** <https://doi.org/10.1002/cne.901640107> | [Google Scholar](#)
- Xu H, Jeong HY, Tremblay R, Rudy B (2013) **Neocortical Somatostatin-Expressing GABAergic Interneurons Disinhibit the Thalamorecipient Layer 4** *Neuron* **77** <https://doi.org/10.1016/j.neuron.2012.11.004> | [Google Scholar](#)
- Yamashita T, Pala A, Pedrido L, Kremer Y, Welker E, Petersen CCH (2013) **Membrane potential dynamics of neocortical projection neurons driving target-specific signals** *Neuron* **80** <https://doi.org/10.1016/j.neuron.2013.10.059> | [Google Scholar](#)
- Yang JW, Prouvot PH, Reyes-Puerta V, Stüttgen MC, Stroh A, Luhmann HJ (2017) **Optogenetic modulation of a minor fraction of parvalbumin-positive interneurons specifically affects spatiotemporal dynamics of spontaneous and sensory-evoked activity in mouse somatosensory cortex in vivo** *Cerebral Cortex* **27** <https://doi.org/10.1093/cercor/bhx261> | [Google Scholar](#)

Young TR, Yamamoto M, Kikuchi SS, Yoshida AC, Abe T, Inoue K, Johansen JP, Benucci A, Yoshimura Y, Shimogori T (2023) **Thalamocortical control of cell-type specificity drives circuits for processing whisker-related information in mouse barrel cortex** *Nat Commun* **14** <https://doi.org/10.1038/s41467-023-41749-x> | [Google Scholar](#)

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Reviewer #1 (Public review):

Summary:

Argunşah et al. describe and investigate the mechanisms underlying the differential response dynamics of barrel vs septal domains of the whisker-related primary somatosensory cortex (S1). Upon repeated stimulation, the authors report that the response ratio between multi- and single-whisker stimulation increases in layer (L) 4 neurons of the septal domain, while remaining constant in barrel L4 neurons. This difference is attributed to the short-term plasticity properties of interneurons, particularly somatostatin-expressing (SST+) neurons. This claim is supported by the increased density of SST+ neurons found in L4 of the septa compared to barrels, along with a stronger response of (L2/3) SST+ neurons to repeated multi- vs single-whisker stimulation. The role of the synaptic protein Elfn1 is then examined. Elfn1 KO mice exhibited little to no functional domain separation between barrel and septa, with no significant difference in single- versus multi-whisker response ratios across barrel and septal domains. Consistently, a decoder trained on WT data fails to generalize to Elfn1 KO responses. Finally, the authors report a relative enrichment of S2- and M1-projecting cell densities in L4 of the septal domain compared to the barrel domain.

Strengths:

This paper describes and aims to study a circuit underlying differential response between barrel columns and septal domains of the primary somatosensory cortex. This work supports the view that barrel and septal domains contribute differently to processing single versus multi-whisker inputs, suggesting that the barrel cortex multiplexes sensory information coming from the whiskers in different domains.

Weaknesses:

While the observed divergence in responses to repeated SWS vs MWS between the barrel and septal domains is intriguing, the presented evidence falls short of demonstrating that short-term plasticity in SST+ neurons critically underpins this difference. The absence of a mechanistic explanation for this observation limits the work's significance. The measurement of SST neurons' response is not specific to a particular domain, and the Elfn1 manipulation does not seem to be specific to either stimulus type or a particular domain. The study's reach is further constrained by the fact that results were obtained in anesthetized animals, which may not generalize to awake states. The statistical analysis appears inappropriate, with the use of repeated independent tests, dramatically boosting the false positive error rate. Furthermore, the manuscript suffers from imprecision; its conclusions are occasionally vague or overstated.

The authors suggest a role for SST+ neurons in the observed divergence in SWS/MWS responses between barrel and septal domains. However, this remains speculative, and some findings appear inconsistent. For instance, the increased response of SST+ neurons to MWS versus SWS is not confined to a specific domain. Why, then, would preferential recruitment of

SST+ neurons lead to divergent dynamics between barrel and septal regions? The higher density of SST+ neurons in septal versus barrel L4 is not a sufficient explanation, particularly since the SWS/MWS response divergence is also observed in layers 2/3, where no difference in SST+ neuron density is found. Moreover, SST+ neuron-mediated inhibition is not necessarily restricted to the layer in which the cell body resides. It remains unclear through which differential microcircuits (barrel vs septum) the enhanced recruitment of SST+ neurons could account for the divergent responses to repeated SWS versus MWS stimulation.

The *Elfn1* KO mouse model seems too unspecific to suggest the role of the short-term plasticity in SST+ neurons in the differential response to repeated SWS vs MWS stimulation across domains. Why would *Elfn1*-dependent short-term plasticity in SST+ neurons be specific to a pathway, or a stimulation type (SWS vs MWS)? Moreover, the authors report that *Elfn1* knockout alters synapses onto VIP+ as well as SST+ neurons (Stachniak et al., 2021; previous version of this paper)-so why attribute the phenotype solely to SST+ circuitry? In fact, the functional distinctions between barrel and septal domains appear largely abolished in the *Elfn1* KO.

<https://doi.org/10.7554/eLife.107099.1.sa3>

Reviewer #2 (Public review):

Summary:

Argunşah and colleagues demonstrate that SST-expressing interneurons are concentrated in the mouse septa and differentially respond to repetitive multi-whisker inputs. Identifying how a specific neuronal phenotype impacts responses is an advance.

Strengths:

- (1) Careful physiological and imaging studies.
- (2) Novel result showing the role of SST+ neurons in shaping responses.
- (3) Good use of a knockout animal to further the main hypothesis.
- (4) Clear analytical techniques.

Weaknesses:

No major weaknesses were identified by this reviewer. Overall I appreciated the paper but feel it overlooked a few issues and had some recommendations on how additional clarifications could strengthen the paper. These include:

- (1) Significant work from Jerry Chen on how S1 neurons that project to M1 versus S2 respond in a variety of behavioral tasks should be included (e.g. PMID: 26098757). Similarly, work from Barry Connor's lab on intracortical versus thalamocortical inputs to SST neurons, as well as excitatory inputs onto these neurons (e.g. PMID: 12815025) should be included.
- (2) Using Layer 2/3 as a proxy to what is happening in layer 4 (~line 234). Given that layer 2/3 cells integrate information from multiple barrels, as well as receiving direct VPM thalamocortical input, and given the time window that is being looked at can receive input from other cortical locations, it is not clear that layer 2/3 is a proxy for what is happening in layer 4.
- (3) Line 267, when discussing distinct temporal response, it is not well defined what this is referring to. Are the neurons no longer showing peaks to whisker stimulation, or are the responses lasting a longer time? It is unclear why PV+ interneurons which may not be

impacted by the *Elfn1* KO and receive strong thalamocortical inputs, are not constraining activity.

(4) Line 585 "the earliest CSD sink was identified as layer 4..." were post-hoc measurements made to determine where the different shank leads were based on the post-hoc histology?

(5) For the retrograde tracing studies, how were the M1 and S2 injections targeted (stereotactically or physiologically)? How was it determined that the injections were in the whisker region (or not)?

(6) Were there any baseline differences in spontaneous activity in the septa versus barrel regions, and did this change in the KO animals?

<https://doi.org/10.7554/eLife.107099.1.sa2>

Reviewer #3 (Public review):

Summary:

This study investigates the functional differences between barrel and septal columns in the mouse somatosensory cortex, focusing on how local inhibitory dynamics, particularly involving *Elfn1*-expressing SST⁺ interneurons, may mediate temporal integration of multi-whisker (MW) stimuli in septa. Using a combination of in vivo multi-unit recordings, calcium imaging, and anatomical tracing, the authors propose that septa integrate MW input in an *Elfn1*-dependent manner, enabling functional segregation from barrel columns.

Strengths:

The core hypothesis is interesting and potentially impactful. While barrels have been extensively characterized, septa remain less understood, especially in mice, and this study's focus on septal integration of MW stimuli offers valuable insights into this underexplored area. If septa indeed act as selective integrators of distributed sensory input, this would add a novel computational role to cortical microcircuits beyond what is currently attributed to barrels alone. The narrative of this paper is intellectually stimulating.

Weaknesses:

The methods used in the current study lack the spatial and cellular resolution needed to conclusively support the central claims. The main physiological findings are based on unsorted multi-unit activity (MUA) recorded via low-channel-count silicon probes. MUA inherently pools signals from multiple neurons across different distances and cell types, making it difficult to assign activity to specific columns (barrel vs. septa) or neuron classes (e.g., SST⁺ vs. excitatory). The recording radius (~50-100 µm or more) and the narrow width of septa (~50-100 µm or less) make it likely that MUA from "septal" electrodes includes spikes from adjacent barrel neurons. The authors do not provide spike sorting, unit isolation, or anatomical validation that would strengthen spatial attribution. Calcium imaging is restricted to SST⁺ and VIP⁺ interneurons in superficial layers (L2/3), while the main MUA recordings are from layer 4, creating a mismatch in laminar relevance.

Furthermore, while the role of *Elfn1* in mediating short-term facilitation is supported by prior studies, no new evidence is presented in this paper to confirm that this synaptic mechanism is indeed disrupted in the knockout mice used here. Additionally, since *Elfn1* is constitutively knocked out from development, the possibility of altered circuit formation—including changes in barrel structure and interneuron distribution, cannot be excluded and is not addressed.

<https://doi.org/10.7554/eLife.107099.1.sa1>

Author response:

Reviewer #1 (Public review):

Summary:

Argunşah et al. describe and investigate the mechanisms underlying the differential response dynamics of barrel vs septa domains of the whisker-related primary somatosensory cortex (S1). Upon repeated stimulation, the authors report that the response ratio between multi- and single-whisker stimulation increases in layer (L) 4 neurons of the septal domain, while remaining constant in barrel L4 neurons. This difference is attributed to the short-term plasticity properties of interneurons, particularly somatostatin-expressing (SST+) neurons. This claim is supported by the increased density of SST+ neurons found in L4 of the septa compared to barrels, along with a stronger response of (L2/3) SST+ neurons to repeated multi- vs single-whisker stimulation. The role of the synaptic protein Elfn1 is then examined. Elfn1 KO mice exhibited little to no functional domain separation between barrel and septa, with no significant difference in single- versus multi-whisker response ratios across barrel and septal domains. Consistently, a decoder trained on WT data fails to generalize to Elfn1 KO responses. Finally, the authors report a relative enrichment of S2- and M1-projecting cell densities in L4 of the septal domain compared to the barrel domain.

Strengths:

This paper describes and aims to study a circuit underlying differential response between barrel columns and septal domains of the primary somatosensory cortex. This work supports the view that barrel and septal domains contribute differently to processing single versus multi-whisker inputs, suggesting that the barrel cortex multiplexes sensory information coming from the whiskers in different domains.

We thank the reviewer for the very neat summary of our findings that barrel cortex multiplexes converging information in separate domains.

Weaknesses:

While the observed divergence in responses to repeated SWS vs MWS between the barrel and septal domains is intriguing, the presented evidence falls short of demonstrating that short-term plasticity in SST+ neurons critically underpins this difference. The absence of a mechanistic explanation for this observation limits the work's significance. The measurement of SST neurons' response is not specific to a particular domain, and the Elfn1 manipulation does not seem to be specific to either stimulus type or a particular domain.

We appreciate the reviewer's perspective. Although further research is needed to understand the circuit mechanisms underlying the observed phenomenon, we believe our data suggest that altering the short-term dynamics of excitatory inputs onto SST neurons reduces the divergent spiking dynamics in barrels versus septa during repetitive single- and multi-whisker stimulation. Future work could examine how SST neurons, whose somata reside in barrels and septa, respond to different whisker stimuli and the circuits in which they are embedded. At this time, however, the authors believe there is no alternative way to test how the short-term dynamics of excitatory inputs onto SST neurons, as a whole, contribute to the temporal aspects of barrel versus septa spiking.

The study's reach is further constrained by the fact that results were obtained in anesthetized animals, which may not generalize to awake states.

We appreciate the reviewer's concern regarding the generalizability of our findings from anesthetized animals to awake states. Anesthesia was employed to ensure precise individual whisker stimulation (and multi-whisker in the same animal), which is challenging in awake rodents due to active whisking. While anesthesia may alter higher-order processing, core mechanisms, such as short and long term plasticity in the barrel cortex, are preserved under anesthesia (Martin-Cortecero et al., 2014; Mégevand et al., 2009).

The statistical analysis appears inappropriate, with the use of repeated independent tests, dramatically boosting the false positive error rate.

Thank you for your feedback on our analysis using independent rank-based tests for each time point in wild-type (WT) animals. To address concerns regarding multiple comparisons and temporal dependencies (for Figure 1F and 4D for now but we will add more in our revision), we performed a repeated measures ANOVA for WT animals (13 Barrel, 8 Septa, 20 time points), which revealed a significant main effect of Condition ($F(1,19) = 16.33$, $p < 0.001$) and a significant Condition-Time interaction ($F(19,361) = 2.37$, $p = 0.001$). Post-hoc tests confirmed significant differences between Barrel and Septa at multiple time points (e.g., $p < 0.0025$ at times 3, 4, 6, 7, 8, 10, 11, 12, 16, 19 after Bonferroni posthoc correction), supporting a differential multi-whisker vs. single-whisker ratio response in WT animals. In contrast, a repeated measures ANOVA for knock-out (KO) animals (11 Barrel, 7 Septa, 20 time points) showed no significant main effect of Condition ($F(1,14) = 0.17$, $p = 0.684$) or Condition-Time interaction ($F(19,266) = 0.73$, $p = 0.791$), indicating that the Barrel-Septa difference observed in WT animals is absent in KO animals.

Furthermore, the manuscript suffers from imprecision; its conclusions are occasionally vague or overstated. The authors suggest a role for SST+ neurons in the observed divergence in SWS/MWS responses between barrel and septal domains. However, this remains speculative, and some findings appear inconsistent. For instance, the increased response of SST+ neurons to MWS versus SWS is not confined to a specific domain. Why, then, would preferential recruitment of SST+ neurons lead to divergent dynamics between barrel and septal regions? The higher density of SST+ neurons in septal versus barrel L4 is not a sufficient explanation, particularly since the SWS/MWS response divergence is also observed in layers 2/3, where no difference in SST+ neuron density is found.

Moreover, SST+ neuron-mediated inhibition is not necessarily restricted to the layer in which the cell body resides. It remains unclear through which differential microcircuits (barrel vs septum) the enhanced recruitment of SST+ neurons could account for the divergent responses to repeated SWS versus MWS stimulation.

We fully appreciate the reviewer's comment. We currently do not provide any evidence on the contribution of SST neurons in the barrels versus septa in layer 4 on the response divergence of spiking observed in SWS versus MWS. We only show that these neurons differentially distribute in the two domains in this layer. It is certainly known that there is molecular and circuit-based diversity of SST-positive neurons in different layers of the cortex, so it is plausible that this includes cells located in the two domains of vS1, something which has not been examined so far. Our data on their distribution are one piece of information that SST neurons may have a differential role in inhibiting barrel stellate cells versus septa ones. Morphological reconstructions of SST neurons in L4 of the somatosensory barrel cortex has shown that their dendrites and axons project locally and may confine to individual domains, even though not specifically examined (Fig. 3 of Scala F et al., 2019). The same study also showed that L4 SST cells receive excitatory input from local stellate cells) and is known that they are also directly excited by thalamocortical fibers (Beierlein et al., 2003; Tan et al., 2008), both of which facilitate.

As shown in our supplementary figure, the divergence is also observed in L2/3 where, as the reviewer also points out, where we do not have a differential distribution of SST cells, at least based on a columnar analysis extending from L4. There are multiple scenarios that could explain this “discrepancy” that one would need to examine further in future studies. One straightforward one is that the divergence in spiking in L2/3 domains may be inherited from L4 domains, where L4 SST act on. Another is that even though L2/3 SST neurons are not biased in their distribution their input-output function is, something which one would need to examine by detailed in vitro electrophysiological and perhaps optogenetic approaches in S1. Despite the distinctive differences that have been found between the L4 circuitry in S1 and V1 (Scala F et al., 2019), recent observations indicate that small but regular patches of V1 marked by the absence of muscarinic receptor 2 (M2) have high temporal acuity (Ji et al., 2015), and selectively receive input from SST interneurons (Meier et al., 2025). Regions lacking M2 have distinct input and output connectivity patterns from those that express M2 (Meier et al., 2021; Burkhalter et al., 2023). These findings, together with ours, suggest that SST cells preferentially innervate and regulate specific domains -columns- in sensory cortices.

Regardless of the mechanism, the *Elfn1* knock-out mouse line almost exclusively affects the incoming excitability onto SST neurons (see also reply to comment below), hence what can be supported by our data is that changing the incoming short-term synaptic plasticity onto these neurons brings the spiking dynamics between barrels and septa closer together.

The Elfn1 KO mouse model seems too unspecific to suggest the role of the short-term plasticity in SST+ neurons in the differential response to repeated SWS vs MWS stimulation across domains. Why would Elfn1-dependent short-term plasticity in SST+ neurons be specific to a pathway, or a stimulation type (SWS vs MWS)? Moreover, the authors report that Elfn1 knockout alters synapses onto VIP+ as well as SST+ neurons (Stachniak et al., 2021; previous version of this paper)-so why attribute the phenotype solely to SST+ circuitry? In fact, the functional distinctions between barrel and septal domains appear largely abolished in the Elfn1 KO.

Previous work by others and us has shown that globally removing *Elfn1* selectively removes a synaptic process from the brain without altering brain anatomy or structure. This allows us to study how the temporal dynamics of inhibition shape activity, as opposed to inhibition from particular cell types. We will nevertheless update the text to discuss more global implications for SST interneuron dynamics and include a reference to VIP interneurons that contain *Elfn1*.

When comparing SWS to MWS, we find that MWS replaces the neighboring excitation which would normally be preferentially removed by short-term plasticity in SST interneurons, thus providing a stable control comparison across animals and genotypes. On average, VIP interneurons failed to show modulation by MWS. We were unable to measure a substantial contribution of VIP cells to this process and also note that the *Elfn1* expressing multipolar neurons comprise only ~5% of VIP neurons (Connor and Peters, 1984; Stachniak et al., 2021), a fraction that may be lost when averaging from 138 VIP cells. Moreover, the effect of *Elfn1* loss on VIP neurons is quite different and marginal compared to that of SST cells, suggesting that the primary impact of *Elfn1* knockout is mediated through SST+ interneuron circuitry. Therefore, even if we cannot rule out that these 5% of VIP neurons contribute to barrel domain segregation, we are of the opinion that their influence would be very limited if any.

Reviewer #2 (Public review):

Summary:

Argunsah and colleagues demonstrate that SST-expressing interneurons are concentrated in the mouse septa and differentially respond to repetitive multi-whisker

inputs. Identifying how a specific neuronal phenotype impacts responses is an advance.

Strengths:

- (1) Careful physiological and imaging studies.*
- (2) Novel result showing the role of SST+ neurons in shaping responses.*
- (3) Good use of a knockout animal to further the main hypothesis.*
- (4) Clear analytical techniques.*

We thank the reviewer for their appreciation of the study.

Weaknesses:

No major weaknesses were identified by this reviewer. Overall, I appreciated the paper but feel it overlooked a few issues and had some recommendations on how additional clarifications could strengthen the paper. These include:

- (1) Significant work from Jerry Chen on how S1 neurons that project to M1 versus S2 respond in a variety of behavioral tasks should be included (e.g. PMID: 26098757). Similarly, work from Barry Connor's lab on intracortical versus thalamocortical inputs to SST neurons, as well as excitatory inputs onto these neurons (e.g. PMID: 12815025) should be included.*

We thank the reviewer for these valuable resources that we overlooked. We will include Chen et al. (2015), Cruikshank et al. (2007) and Gibson et al. (1999) to contextualize S1 projections and SST+ inputs, strengthening the study's foundation as well as Beierlein et al. (2003) which nicely show both local and thalamocortical facilitation of excitatory inputs onto L4 SST neurons, in contrast to PV cells. The paper also shows the gradual recruitment of SST neurons by thalamocortical inputs to provide feed-forward inhibition onto stellate cells (regular spiking) of the barrel cortex L4 in rat.

- (2) Using Layer 2/3 as a proxy to what is happening in layer 4 (~line 234). Given that layer 2/3 cells integrate information from multiple barrels, as well as receiving direct VPM thalamocortical input, and given the time window that is being looked at can receive input from other cortical locations, it is not clear that layer 2/3 is a proxy for what is happening in layer 4.*

We agree with the reviewer that what we observe in L2/3 is not necessarily what is taking place in L4 SST-positive cells. The data on L2/3 was included to show that these cells, as a population, can show divergent responses when it comes to SWS vs MWS, which is not seen in L2/3 VIP neurons. Regardless of the mechanisms underlying it, our overall data support that SST-positive neurons can change their activation based on the type of whisker stimulus and when the excitatory input dynamics onto these neurons change due to the removal of *Elfn1* the recruitment of barrels vs septa spiking changes at the temporal domain. Having said that, the data shown in Supplementary Figure 3 on the response properties of L2/3 neurons above the septa vs above the barrels (one would say in the respective columns) do show the same divergence as in L4. This suggests that a circuit motif may exist that is common to both layers, involving SST neurons that sit in L4, L5 or even L2/3. This implies that despite the differences in the distribution of SST neurons in septa vs barrels of L4 there is an unidentified input-output spatial connectivity motif that engages in both L2/3 and L4. Please also see our response to a similar point raised by reviewer 1.

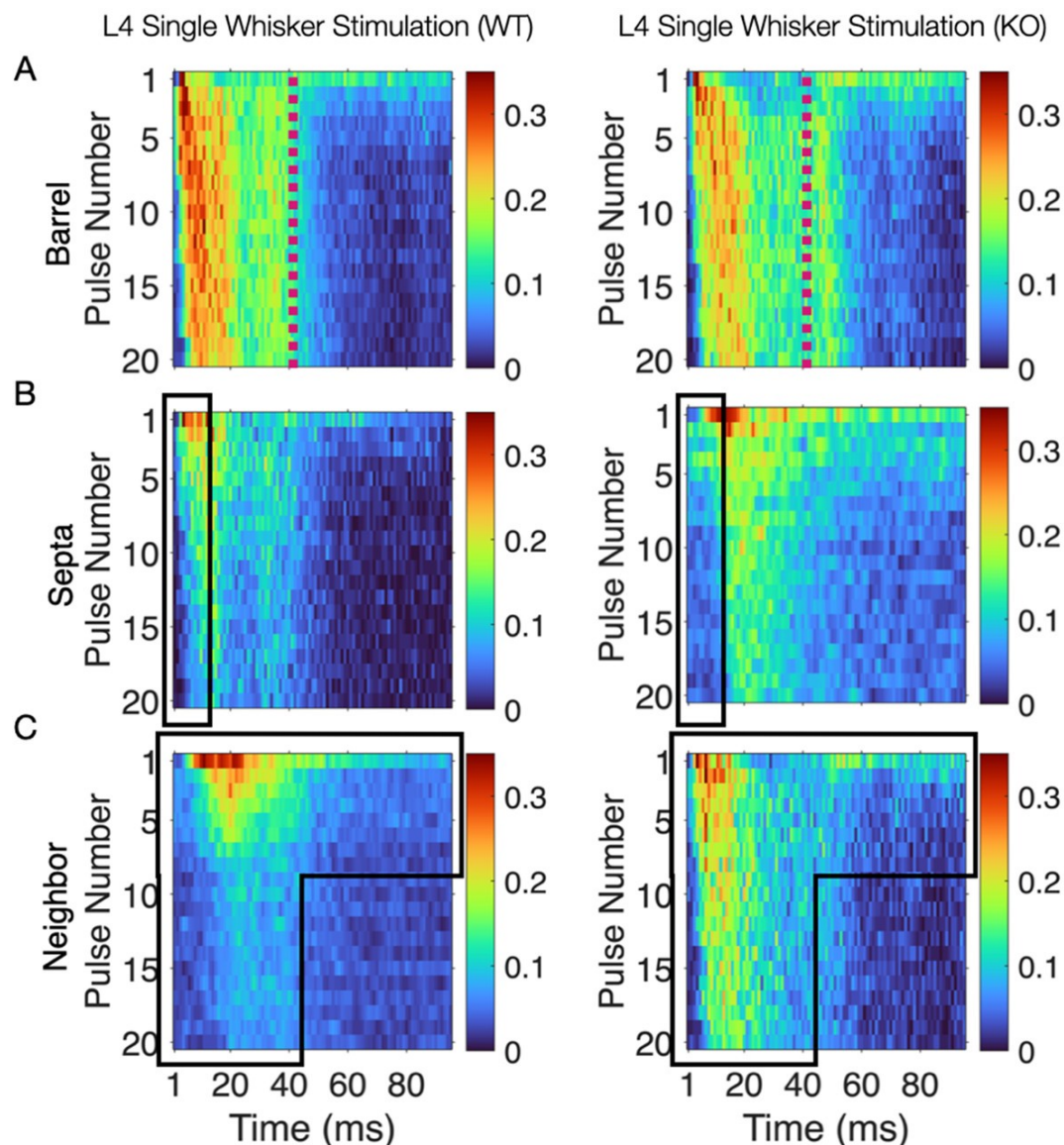
(3) Line 267, when discussing distinct temporal response, it is not well defined what this is referring to. Are the neurons no longer showing peaks to whisker stimulation, or are the responses lasting a longer time? It is unclear why PV+ interneurons which may not be impacted by the Elfn1 KO and receive strong thalamocortical inputs, are not constraining activity.

We thank the reviewer for their comment and will clarify the statement.

This convergence of response profiles was further clear in stimulus-aligned stacked images, where the emergent differences between barrels and septa under SWS were largely abolished in the KO (Figure 4B). A distinction between directly stimulated barrels and neighboring barrels persisted in the KO. In addition, the initial response continued to differ between barrel and septa and also septa and neighbor (Figure 4B). This initial stimulus selectivity potentially represents distinct feedforward thalamocortical activity, which includes PV+ interneuron recruitment that is not directly impacted by the Elfn1 KO (Sun et al., 2006; Tan et al., 2008). PV+ cells are strongly excited by thalamocortical inputs, but these exhibit short-term depression, as does their output, contrasting with the sustained facilitation observed in SST+ neurons. These findings suggest that in WT animals, activity spillover from principal barrels is normally constrained by the progressive engagement of SST+ interneurons in septal regions, driven by Elfn1-dependent facilitation at their excitatory synapses. In the absence of Elfn1, this local inhibitory mechanism is disrupted, leading to longer responses in barrels, delayed but stronger responses in septa, and persistently stronger responses in unstimulated neighbors, resulting in a loss of distinction between the responses of barrel and septa domains that normally diverge over time (see Author response image 1 below).

Author response image 1.

A) Barrel responses are longer following whisker stimulation in KO. B) Septal responses are slightly delayed but stronger in KO. C) Unstimulated neighbors show longer persistent responses in KO.



(4) Line 585 "the earliest CSD sink was identified as layer 4..." were post-hoc measurements made to determine where the different shank leads were based on the post-hoc histology?

Post hoc histology was performed on plane-aligned brain sections which would allow us to detect barrels and septa, so as to confirm the insertion domains of each recorded shank. Layer specificity of each electrode therefore could not be confirmed by histology as we did not have coronal sections in which to measure electrode depth.

(5) For the retrograde tracing studies, how were the M1 and S2 injections targeted (stereotaxically or physiologically)? How was it determined that the injections were in the whisker region (or not)?

During the retrograde virus injection, the location of M1 and S2 injections was determined by stereotaxic coordinates (Yamashita et al., 2018). After acquiring the light-sheet images, we were able to post hoc examine the injection site in 3D and confirm that the injections were successful in targeting the regions intended. Although it would have been informative to do

so, we did not functionally determine the whisker-related M1 and whisker-related S2 region in this experiment.

(6) *Were there any baseline differences in spontaneous activity in the septa versus barrel regions, and did this change in the KO animals?*

Thank you for this interesting question. Our previous study found that there was a reduction in baseline activity in L4 barrel cortex of KO animals at postnatal day (P)12, but no differences were found at P21 (Stachniak et al., 2023).

Reviewer #3 (Public review):

Summary:

*This study investigates the functional differences between barrel and septal columns in the mouse somatosensory cortex, focusing on how local inhibitory dynamics, particularly involving *Elfn1*-expressing *SST*⁺ interneurons, may mediate temporal integration of multi-whisker (MW) stimuli in septa. Using a combination of in vivo multi-unit recordings, calcium imaging, and anatomical tracing, the authors propose that septa integrate MW input in an *Elfn1*-dependent manner, enabling functional segregation from barrel columns.*

Strengths:

The core hypothesis is interesting and potentially impactful. While barrels have been extensively characterized, septa remain less understood, especially in mice, and this study's focus on septal integration of MW stimuli offers valuable insights into this underexplored area. If septa indeed act as selective integrators of distributed sensory input, this would add a novel computational role to cortical microcircuits beyond what is currently attributed to barrels alone. The narrative of this paper is intellectually stimulating.

We thank the reviewer for finding the study intellectually stimulating.

Weaknesses:

*The methods used in the current study lack the spatial and cellular resolution needed to conclusively support the central claims. The main physiological findings are based on unsorted multi-unit activity (MUA) recorded via low-channel-count silicon probes. MUA inherently pools signals from multiple neurons across different distances and cell types, making it difficult to assign activity to specific columns (barrel vs. septa) or neuron classes (e.g., *SST*⁺ vs. excitatory).*

The recording radius (~50-100 μ m or more) and the narrow width of septa (~50-100 μ m or less) make it likely that MUA from "septal" electrodes includes spikes from adjacent barrel neurons.

*The authors do not provide spike sorting, unit isolation, or anatomical validation that would strengthen spatial attribution. Calcium imaging is restricted to *SST*⁺ and *VIP*⁺ interneurons in superficial layers (L2/3), while the main MUA recordings are from layer 4, creating a mismatch in laminar relevance.*

We thank the reviewer for pointing out the possibility of contamination in septal electrodes. Importantly, it may not have been highlighted, although reported in the methods, but we used an extremely high threshold (7.5 std, in methods, line 583) for spike detection in order to overcome the issue raised here, which restricts such spatial contaminations. Since the spike

amplitude decays rapidly with distance, at high thresholds, only nearby neurons contribute to our analysis, potentially one or two. We believe that this approach provides a very close approximation of single unit activity (SUA) in our reported data. We will include a sentence earlier in the manuscript to make this explicit and prevent further confusion.

Regarding the point on calcium imaging being performed on L2/3 SST and VIP cells instead of L4. Both reviewer 1 and 2 brought up the same issue and we responded as follows. As shown in our supplementary figure, the divergence is also observed in L2/3 where we do not have a differential distribution of SST cells, at least based on a columnar analysis extending from L4. There are multiple scenarios that could explain this “discrepancy” that one would need to examine further in future studies. One straightforward one is that the divergence in spiking in L2/3 domains may be inherited from L4 domains, where L4 SST act on. Another is that even though L2/3 SST neurons are not biased in their distribution their input-output function is, something which one would need to examine by detailed in vitro electrophysiological and perhaps optogenetic approaches in S1. Despite the distinctive differences that have been found between the L4 circuitry in S1 and V1 (Scala F et al., 2019), recent observations indicate that small but regular patches of V1 marked by the absence of muscarinic receptor 2 (M2) have high temporal acuity (Ji et al., 2015), and selectively receive input from SST interneurons (Meier et al., 2025). Regions lacking M2 have distinct input and output connectivity patterns from those that express M2 (Meier et al., 2021; Burkhalter et al., 2023). These findings, together with ours, suggest that SST cells preferentially innervate and regulate specific domains -columns- in sensory cortices.

Furthermore, while the role of Elfn1 in mediating short-term facilitation is supported by prior studies, no new evidence is presented in this paper to confirm that this synaptic mechanism is indeed disrupted in the knockout mice used here.

We thank Reviewer #3 for noting the absence of new evidence confirming Elfn1’s disruption of short-term facilitation in our knockout mice. We acknowledge that our study relies on previously strong published data demonstrating that Elfn1 mediates short-term synaptic facilitation of excitatory inputs onto SST+ interneurons (Sylwestrak and Ghosh, 2012; Tomioka et al., 2014; Stachniak et al., 2019, 2023). These studies consistently show that Elfn1 knockout abolishes facilitation in SST+ synapses, leading to altered temporal dynamics, which we hypothesize underlies the observed loss of barrel-septa response divergence in our Elfn1 KO mice (Figure 4). Nevertheless, to address the point raised, we will clarify in the revised manuscript (around lines 245-247 and 271-272) that our conclusions are based on these established findings, stating: “Building on prior evidence that Elfn1 knockout disrupts short-term facilitation in SST+ interneurons (Sylwestrak and Ghosh, 2012; Tomioka et al., 2014; Stachniak et al., 2019, 2023), we attribute the abolished barrel-septa divergence in Elfn1 KO mice to altered SST+ synaptic dynamics, though direct synaptic measurements were not performed here.”

Additionally, since Elfn1 is constitutively knocked out from development, the possibility of altered circuit formation-including changes in barrel structure and interneuron distribution, cannot be excluded and is not addressed.

We thank Reviewer #3 for raising the valid concern that constitutive Elfn1 knockout could potentially alter circuit formation, including barrel structure and interneuron distribution. To address this, we will clarify in the revised manuscript (around line ~271 and in the Discussion) that in our previous studies that included both whole-cell patch-clamp in acute brain slices ranging from postnatal day 11 to 22 (P11 - P21) and in vivo recordings from barrel cortex at P12 and P21, we saw no gross abnormalities in barrel structure, with Layer 4 barrels maintaining their characteristic size and organization, consistent with wild-type (WT) mice (Stachniak et al., 2019, 2023). While we cannot fully exclude subtle developmental

changes, prior studies indicate that *Elfn1* primarily modulates synaptic function rather than cortical cytoarchitecture (Tomioka et al., 2014). *Elfn1* KO mice show no gross morphological or connectivity differences and the pattern and abundance of *Elfn1* expressing cells (assessed by LacZ knock in) appears normal (Dolan and Mitchell, 2013).

We will add the following to the Discussion: “Although *Elfn1* is constitutively knocked out, we find here and in previous studies that barrel structure is preserved (Stachniak et al., 2019, 2023). Further, the distribution of *Elfn1* expressing interneurons is not different in KO mice, suggesting minimal developmental disruption (Dolan and Mitchell, 2013). Nonetheless, we acknowledge that subtle circuit changes cannot be ruled out without the usage of time-dependent conditional knockout of the gene.”

References

- (1) Beierlein, M., Gibson, J. R. & Connors, B. W. (2003). Two dynamically distinct inhibitory networks in layer 4 of the neocortex. *J. Neurophysiol.* 90, 2987–3000.
- (2) Burkhalter, A., D’Souza, R. D. & Ji, W. (2023). Integration of feedforward and feedback information streams in the modular architecture of mouse visual cortex. *Annu. Rev. Neurosci.* 46, 259–280.
- (3) Chen, J. L., Margolis, D. J., Stankov, A., Sumanovski, L. T., Schneider, B. L. & Helmchen, F. (2015). Pathway-specific reorganization of projection neurons in somatosensory cortex during learning. *Nat. Neurosci.* 18, 1101–1108.
- (4) Connor, J. R. & Peters, A. (1984). Vasoactive intestinal polypeptide-immunoreactive neurons in rat visual cortex. *Neuroscience* 12, 1027–1044.
- (5) Cruikshank, S. J., Lewis, T. J. & Connors, B. W. (2007). Synaptic basis for intense thalamocortical activation of feedforward inhibitory cells in neocortex. *Nat. Neurosci.* 10, 462–468.
- (6) Dolan, J. & Mitchell, K. J. (2013). Mutation of *Elfn1* in mice causes seizures and hyperactivity. *PLoS One* 8, e80491.
- (7) Gibson, J. R., Beierlein, M. & Connors, B. W. (1999). Two networks of electrically coupled inhibitory neurons in neocortex. *Nature* 402, 75–79.
- (8) Ji, W., Gămănuț, R., Bista, P., D’Souza, R. D., Wang, Q. & Burkhalter, A. (2015). Modularity in the organization of mouse primary visual cortex. *Neuron* 87, 632–643.
- (9) Martin-Cortecero, J. & Nuñez, A. (2014). Tactile response adaptation to whisker stimulation in the lemniscal somatosensory pathway of rats. *Brain Res.* 1591, 27–37.
- (10) Mégevand, P., Troncoso, E., Quairiaux, C., Muller, D., Michel, C. M. & Kiss, J. Z. (2009). Long-term plasticity in mouse sensorimotor circuits after rhythmic whisker stimulation. *J. Neurosci.* 29, 5326–5335.
- (11) Meier, A. M., Wang, Q., Ji, W., Ganachaud, J. & Burkhalter, A. (2021). Modular network between postrhinal visual cortex, amygdala, and entorhinal cortex. *J. Neurosci.* 41, 4809–4825.
- (12) Meier, A. M., D’Souza, R. D., Ji, W., Han, E. B. & Burkhalter, A. (2025). Interdigitating modules for visual processing during locomotion and rest in mouse V1. *bioRxiv* 2025.02.21.639505.
- (13) Scala, F., Kobak, D., Shan, S., Bernaerts, Y., Laturnus, S., Cadwell, C. R., Hartmanis, L., Froudarakis, E., Castro, J. R., Tan, Z. H., et al. (2019). Layer 4 of mouse neocortex differs in cell types and circuit organization between sensory areas. *Nat. Commun.* 10, 4174.

- (14) Stachniak, T. J., Sylwestrak, E. L., Scheiffele, P., Hall, B. J. & Ghosh, A. (2019). Elfn1-induced constitutive activation of mGluR7 determines frequency-dependent recruitment of somatostatin interneurons. *J. Neurosci.* 39, 4461–4475.
- (15) Stachniak, T. J., Kastli, R., Hanley, O., Argunsah, A. Ö., van der Valk, E. G. T., Kanatouris, G. & Karayannis, T. (2021). Postmitotic Prox1 expression controls the final specification of cortical VIP interneuron subtypes. *J. Neurosci.* 41, 8150–8166.
- (16) Stachniak, T. J., Argunsah, A. Ö., Yang, J. W., Cai, L. & Karayannis, T. (2023). Presynaptic kainate receptors onto somatostatin interneurons are recruited by activity throughout development and contribute to cortical sensory adaptation. *J. Neurosci.* 43, 7101–7118.
- (17) Sun, Q.-Q., Huguenard, J. R. & Prince, D. A. (2006). Barrel cortex microcircuits: Thalamocortical feedforward inhibition in spiny stellate cells is mediated by a small number of fast-spiking interneurons. *J. Neurosci.* 26, 1219–1230.
- (18) Sylwestrak, E. L. & Ghosh, A. (2012). Elfn1 regulates target-specific release probability at CA1-interneuron synapses. *Science* 338, 536–540.
- (19) Tan, Z., Hu, H., Huang, Z. J. & Agmon, A. (2008). Robust but delayed thalamocortical activation of dendritic-targeting inhibitory interneurons. *Proc. Natl. Acad. Sci. USA* 105, 2187–2192.
- (20) Tomioka, N. H., Yasuda, H., Miyamoto, H., Hatayama, M., Morimura, N., Matsumoto, Y., Suzuki, T., Odagawa, M., Odaka, Y. S., Iwayama, Y., et al. (2014). Elfn1 recruits presynaptic mGluR7 in trans and its loss results in seizures. *Nat. Commun.* 5, 4501.
- (21) Yamashita, T., Vavladeli, A., Pala, A., Galan, K., Crochet, S., Petersen, S. S. & Petersen, C. C. (2018). Diverse long-range axonal projections of excitatory layer 2/3 neurons in mouse barrel cortex. *Front. Neuroanat.* 12, 33.

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